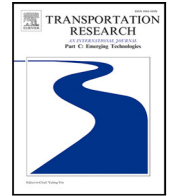




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Modeling car-following behaviors considering driver heterogeneity: A multi-regime stochastic framework

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ABSTRACT

Car-following behavior exhibits stochastic characteristics influenced by the inherent randomness of human drivers. Stochastic models have been extensively developed to capture the probabilistic nature of car-following behavior dynamics. However, the time-varying nature driven by driver heterogeneity has not been adequately studied. To this end, this paper proposes a stochastic modeling framework that incorporates multi-regime car-following models with a Bayesian calibration approach to account for the driver heterogeneity in human car-following behaviors. More specifically, our framework employs a K-means clustering algorithm to categorize human drivers into three driving styles and leverages a hierarchical Dirichlet process-hidden semi-Markov model (HDP-HSMM) to segment car-following sequences into diverse driving regimes, thereby capturing driver heterogeneity. According to the segmented driving regimes, three distinct hierarchies of multi-regime Bayesian intelligent driver models (denoted pooled, hierarchical, and unpooled B-IDM) are developed to capture the time-varying nature of car-following behaviors across diverse driving regimes. These models are well-calibrated using a Bayesian approach with car-following trajectory data extracted from the Waymo open motion dataset and Lyft level-5 dataset. Deterministic and stochastic simulations are performed to evaluate the effectiveness of the proposed stochastic modeling framework. The experimental results demonstrate significant differences in car-following behaviors across various driving styles and driving regimes. The proposed framework effectively represents these heterogeneous car-following behaviors through the developed multi-regime hierarchical B-IDM with time-varying parameters. Additionally, the stochastic simulation achieves a more accurate representation than the deterministic simulation in replicating the dynamics of human car-following behaviors.

1. Introduction

Car-following behavior constitutes one of the primary driving tasks that drivers consistently perform. Thus, modeling human car-following behavior is fundamental for accurately simulating traffic flow dynamics and evaluating traffic operations, safety, and efficiency. However, the complexity and dynamics of human driving behaviors can lead to various traffic issues, such as traffic oscillations, and reduction in traffic capacity (Jia et al., 2014; Sun et al., 2018; Ngoduy, 2021). Therefore, it is essential to provide an accurate description of heterogeneous car-following behaviors (Ding et al., 2022; Rao et al., 2023). This enables precise simulation of microscopic driving behaviors, which in turn reflect real-world traffic flow dynamics.

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Car-following models are designed to describe the longitudinal movement of a following vehicle in response to the corresponding leading vehicle. Over the past decades, numerous car-following models have been proposed to approximately simulate the dynamic human car-following behaviors in traffic flow (Saifuzzaman and Zheng, 2014). These models are generally classified as mathematical models, utilizing mathematical formulations with a finite set of parameters to describe the vehicle movements, akin to a driving machine. Additionally, each parameter has a distinct physical meaning, allowing these models to capture heterogeneous car-following behaviors of individual drivers through appropriate parameter settings. For instance, the intelligent driver model (IDM) (Treiber et al., 2000), a popular model, incorporates desired space headway and desired speed to compute the acceleration of the following vehicle. The different values of desired space headway and speed can effectively capture inter-driver heterogeneity (Treiber and Helbing, 2003; Zhu et al., 2018b).

Recently, data-driven models have attracted much attention for modeling car-following behaviors. These models are capable of capturing complex, non-linear relationships from historical large-scale datasets and accounting for driver heterogeneity in car-following behaviors, facilitating the prediction of individual driver behavior. The commonly used models include K-nearest neighbor (KNN) algorithm (He et al., 2015), long short-term memory (LSTM) network (Wang et al., 2017), Bayesian Gaussian mixture model (Chen et al., 2023), general reinforcement learning-based methods (Zhu et al., 2018b; Wen et al., 2023). For instance, Morton et al. (2016) utilized an LSTM network to model car-following behavior, providing accurate predictions of acceleration distribution. Zhu et al. (2020) developed a deep reinforcement learning-based car-following control strategy that effectively balances safety, efficiency, and ride comfort in autonomous driving. However, despite significant improvements in modeling capabilities, the interpretability of such models remains limited due to their inherently black-box nature.

Furthermore, hybrid models that integrate mathematical and data-driven approaches have been increasingly utilized to characterize human car-following behaviors (Jiang et al., 2024). For example, Mo et al. (2021) designed a physics-informed car-following architecture that leverages the advantages of interpretable mathematical models and the generalizability of deep learning methods. The results demonstrate that the physics-informed deep learning model outperforms the physics-uninformed deep learning model. Pan et al. (2023) proposed a hybrid model that integrates the IDM and bidirectional LSTM. The framework preserves the controllability and safety of mathematical models while leveraging the predictive accuracy of data-driven models. Moreover, Li et al. (2025a) developed an artificial intelligence-driven model that employs a neural network for symbolic regression (Sci-CFM), generating a tractable mathematical expression to investigate the underlying mechanisms of car-following behaviors. Besides, some reinforcement learning-based methods, known as physics-informed reinforcement learning, incorporate physical knowledge (e.g., driving behavior characteristics, car-following model parameters, etc.) into the reward function to improve model performance and interpretability (Liu et al., 2023; Wang et al., 2025). For example, Hart et al. (2024) developed a robust reinforcement learning-based car-following model for safety-critical scenarios, in which driving behavior characteristics are embedded in the reward function to enhance policy realism, stability, and safety performance.

Recent studies have identified probabilistic approaches as a promising development in stochastic car-following behavior modeling (Lee et al., 2019; Luo et al., 2023). The stochastic models incorporate a random component to account for the dynamic and uncertainty of human driving behavior. For example, Laval et al. (2014) extended a stochastic desired acceleration model, building upon the Newell framework and incorporating white noise into the desired acceleration. Similarly, Ngoduy et al. (2019) developed a stochastic model that introduces a random term into the deterministic formula to describe the stochastic driving characteristics in different speed regimes. Tian et al. (2021) proposed a stochastic Newell model to simulate microscopic car-following behaviors accurately. To consider the changes in moment-to-moment behaviors, Zhang et al. (2022) proposed a generative model that integrates a neural process and IDM with time-varying parameters to capture the dynamic car-following behaviors. However, a driver may change his/her driving style under different driving conditions (driving regimes), rather than shifting from moment to moment. Hence, capturing temporal changes in model parameters across different driving regimes, rather than on a moment-to-moment basis, is more suitable for representing intra-driver heterogeneity in car-following behaviors (Zaky and Gomaa, 2014; Taylor et al., 2015). Kang et al. (2023) divided a car-following sequence into two or three driving regimes, considering the surrounding vehicle's information, and added random coefficients to describe heterogeneous car-following behaviors. Overall, stochastic models provide a robust framework that integrates various machine learning techniques with dynamic adjustments to capture diverse and complex car-following behaviors.

In summary, each model has its limitations. Mathematical models are transparent and interpretable but often lack the flexibility to capture the complexity and heterogeneity of car-following behaviors. Data-driven models can effectively capture the complexities and variations, but often lack interpretability. Stochastic models exhibit strong potential in representing the inherent heterogeneity of car-following behaviors. However, most existing stochastic approaches fail to account for the time-varying characteristics. To address this issue, this study proposes a stochastic modeling framework that integrates multi-regime car-following models with a Bayesian calibration approach to account for driver heterogeneity in human car-following behaviors. The framework employs a K-means clustering algorithm and a hierarchical Dirichlet process-hidden semi-Markov model (HDP-HSMM) (Johnson and Willsky, 2013) to classify driving styles and segment driving regimes, respectively. This enables the capture of both inter-driver and intra-driver heterogeneity. Subsequently, a Bayesian approach (Zhang and Sun, 2024) is employed to calibrate the multi-regime Bayesian IDM, leveraging the advantages of stochastic processes and time-varying parameters to improve model accuracy. The contributions are summarized as follows:

- A multi-regime stochastic framework is proposed to model heterogeneous human car-following behaviors, accounting for both inter-driver and intra-driver heterogeneity through driving style classification and driving regime segmentation. This framework offers a precise and comprehensive representation of heterogeneous human car-following behaviors.

- Three distinct hierarchies of multi-regime Bayesian intelligent driver models (i.e., pooled, hierarchical, and unpooled B-IDM) are developed and calibrated using a Bayesian approach. In particular, the hierarchical B-IDM provides distinct model parameters to characterize heterogeneous car-following behaviors at the individual level (driving regime), while simultaneously accounting for diverse driving styles at the population level. Consequently, heterogeneous driving behaviors within the same driver can be generated through the well-calibrated posterior distributions of model parameters.
- The stochastic simulation is performed, drawing on the generative process inherent in the Bayesian calibration approach. As a result, the stochastic simulation provides more realistic and representative outcomes than the deterministic simulation.

The rest of the paper is organized as follows. Section 2 introduces related works about modeling heterogeneity in car-following behaviors and car-following model calibration approaches. Section 3 introduces the proposed car-following behavior modeling framework, which includes the driving regime segmentation approach and multi-regime car-following models, that is, three distinct hierarchies of B-IDM. Section 4 describes the data preparation process, including the use of the Waymo open motion dataset and the Lyft level-5 dataset. Section 5 discusses the experimental and simulation results, and the main conclusions are briefly summarized in Section 6.

2. Related works

2.1. Modeling heterogeneity in car-following behaviors

The heterogeneity in car-following behaviors is divided into inter-driver heterogeneity and intra-driver heterogeneity. Inter-driver heterogeneity is typically reflected in the diverse driving styles exhibited by different drivers, whereas intra-driver heterogeneity is characterized by changes in driving styles within the same driver (Zhang et al., 2022; Huang et al., 2024). Previous studies have proposed various approaches to capture driver heterogeneity when modeling human car-following behaviors.

Some scholars have attempted to calibrate different model parameter sets using trajectory data from diverse drivers (Ossen and Hoogendoorn, 2011; James et al., 2018). Each model parameter set presents a distinct driving style, capturing the inter-driver heterogeneity. For instance, James et al. (2018) divided the trajectory data into three groups and calibrated a model parameter set for each group. The differences in the parameter set provide evidence of heterogeneity. Chen et al. (2020) also employed various clustering methods to categorize drivers into normal, aggressive, and timid styles, highlighting the long-term driving characteristics (i.e., inter-driver heterogeneity) in car-following. Besides, Kim et al. (2013) introduced a random coefficient to capture the heterogeneity among drivers, reflecting the diverse ways in which they respond to stimuli when modeling human car-following behaviors. Huang et al. (2024) developed a fog-related IDM, that incorporates unobserved driver heterogeneity to replicate driver behavioral patterns. These findings suggest that the distributions of model parameters are capable of capturing inter-driver heterogeneity when modeling car-following behaviors with mathematical models.

The intra-driver heterogeneity is generally presented in the variations in driving behavior over time. One approach to capture this heterogeneity is through stochastic modeling that incorporates temporal features. For instance, Zhang and Sun (2024) developed a memory-augmented IDM (MA-IDM) using a Gaussian process to account for the temporally correlated errors. Later, Zhang et al. (2024) proposed a car-following model that incorporates an autoregressive process to capture dynamic driver behaviors. Another approach aims to calibrate distinct model parameters for different driving modes (e.g., acceleration, deceleration) within the same trajectory. This approach effectively accounts for the time-varying nature of car-following behavior. Taylor et al. (2015) applied dynamic time warping (DTW) to segment the car-following sequence into distinct driving regimes and calibrated the Helly model parameters for each regime, effectively capturing intra-driver heterogeneity. Xu et al. (2018) incorporated a stimulus-responsive model with a dynamic Bayesian network to account for the intra-driver heterogeneity across distinct driving regimes. Similarly, Zaky and Goma (2014) employed a Markov switching model to segment driving regimes, allowing time-varying model parameters to represent heterogeneity in car-following behaviors. Besides, various approaches have been proposed to segment driving regimes (also referred to as interaction patterns), such as deep sparse autoencoder (Liu et al., 2014), variational autoencoder (Kang et al., 2023), hidden Markov model and its variants (Wang et al., 2018; Zou et al., 2022, 2025).

2.2. Car-following model calibration

Appropriate parameters of car-following models are fundamental for accurately describing complex driving behaviors. Generally, these parameters are estimated by optimizing specific objective functions using car-following trajectory data. This process is regarded as calibration. Previous studies have proposed various calibration methods, including heuristic algorithms, maximum likelihood estimation (MLE) methods, and Bayesian approaches.

The first type is heuristic algorithms, represented by the genetic algorithm (Sharma et al., 2019). For example, Zhu et al. (2018a) employed the Shanghai naturalistic driving dataset to calibrate five mathematical models, including the Gipps model, IDM, full velocity difference (FVD) model, etc., utilizing a genetic algorithm. Punzo et al. (2021) compared 29 objective functions, derived from combinations of 7 goodness-of-fit functions and 3 measures of performance, in car-following model calibration using the genetic algorithm. To calibrate stochastic car-following models, Zhou et al. (2025) developed a multiple runs minimum error method based on a genetic algorithm to capture the inherent stochasticity of car-following behavior and reduce estimation errors. However, the calibration results are sensitive to data quality, as well as the choice of goodness-of-fit function and measure of performance (Li et al., 2025b). Additionally, the calibrated parameters may be suboptimal if the optimization algorithm converges to local optima.

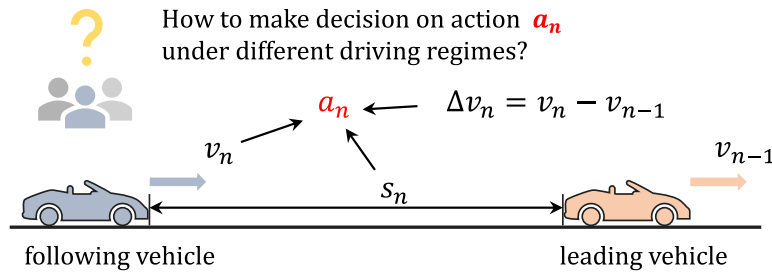


Fig. 1. Illustration of the car-following decision.

The MLE method is another method that is aimed at estimating optimal model parameters. Hoogendoorn and Hoogendoorn (2010) first applied the MLE method to calibrate the parameters of car-following models. Treiber and Kesting (2013) further investigated the MLE method on car-following model calibration and validation. Xu and Laval (2020) utilized the MLE method to calibrate the parameters of the two-regime stochastic car-following models. Similarly, Lee et al. (2019) also employed the MLE method to calibrate the hybrid model that integrates deep learning with a stochastic car-following model.

Unlike the MLE method, the Bayesian approach provides an alternative framework for estimating parameter distribution, often utilizing sampling-based algorithms to derive posterior distributions. Rahman et al. (2015) used the Bayesian approach to calibrate car-following models. The results demonstrated that the Bayesian approach obtained better performance than the optimization algorithm. Jiang et al. (2025) proposed the use of approximate Bayesian computation for stochastic calibration of car-following models, which has yielded satisfactory results. Recently, Zhang et al. (2024) utilized a Bayesian calibration approach to calibrate the dynamic IDM with an autoregressive process. The study further extended the calibration process to account for both speed and position. In summary, the Bayesian approach offers posterior distributions of model parameters rather than a single set of calibrated parameter values, demonstrating superior calibration performance for stochastic car-following models. Consequently, this study adopts the Bayesian approach for parameter estimation.

3. Methodology

3.1. Problem statement

Human car-following behavior refers to the driver's continuous adjustment of speed and position in response to the leading vehicle, according to the perceived traffic environment, which is characterized by spacing (s), speed (v), and speed difference (Δv). However, due to the complexity and uncertainty of human driving behavior, different drivers may take various actions in response to the same perceived information, and even the same driver may respond differently depending on the specific traffic conditions. These phenomena are referred to as inter-driver heterogeneity and intra-driver heterogeneity in car-following behaviors, as noted in previous studies (Chen et al., 2020; Zhang et al., 2022; Wang et al., 2025). Therefore, as shown in Fig. 1, the actions (i.e., acceleration a) a driver takes are influenced by both his/her driving style and the specific driving regime in which he/she is operating.

To account for inter-driver and intra-driver heterogeneity in human car-following behaviors, this study develops a multi-regime stochastic modeling framework. The framework employs a K-means clustering method to group three distinct driving styles, addressing inter-driver heterogeneity, and utilizes an HDP-HSMM for segmenting driving regimes, thereby capturing intra-driver heterogeneity. Consequently, multi-regime car-following models are developed and calibrated, incorporating time-varying model parameters to effectively represent the heterogeneous car-following behaviors.

3.2. HDP-HSMM for driving regime segmentation

The hidden Markov model (HMM) is a probabilistic statistical model that describes a Markov process with hidden states. It generates an unobserved hidden state sequence via a Markov chain, and the observed sequence is generated based on the hidden state sequence. HMM has been widely used to solve sequential problems (e.g., speech recognition, natural language processing, time series problems, etc.). However, the model suffers from two key limitations. First, it assumes a geometric distribution for state durations, which is often unrealistic for real-world applications. Second, it requires a predefined number of hidden states, increasing model complexity and reducing flexibility (Johnson and Willsky, 2013). The Hidden semi-Markov model (HSMM) is an extension of HMM that introduces a state duration distribution to address the first limitation of HMM. The fundamental difference between HMM and HSMM is that HSMM generates a sequence of observations from a single hidden state over a specified duration. Moreover, the hierarchical Dirichlet process (HDP) provides a flexible framework for inferring infinite states from data. Building on this, the HDP-HSMM was proposed by Johnson and Willsky (2013) to allow the number of hidden states to be determined automatically, and to accommodate general, non-geometric duration distributions.

The graphical model of HDP-HSMM is illustrated in Fig. 2. There are four distinct sequences: the super-state sequence $\{z_r\}_{r=1}^R$, the duration sequence $\{d_r\}_{r=1}^R$, the label sequence $\{x_t\}_{t=1}^T$, and the observation sequence $\{y_t\}_{t=1}^T$, where r and t denote the state index

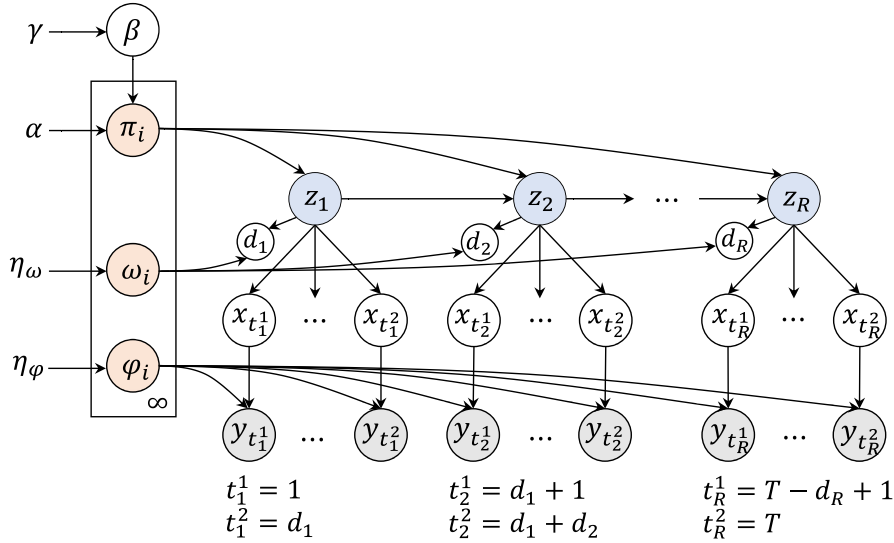


Fig. 2. Graphical model of HDP-HSMM. For each upper-state z_r (with $r = 1, 2, \dots, R$), the associated time segment is denoted by t_r , which spans from the start time index t_r^1 to the end time index t_r^2 .

and time index, respectively. Specifically, z_r and d_r denote the super-state r and its corresponding duration, while x_t and y_t represent the label and observation at time t , respectively. Note that the number of states (or driving regimes) is a latent random variable, while the observation sequence length is observed. The generative process can be summarized as follows:

$$\beta \sim \text{GEM}(\gamma) \quad (1)$$

$$\pi_i \sim \text{DP}(\alpha, \beta), i = 1, 2, \dots, \quad (2)$$

$$(\varphi_i, \vartheta_i) \sim (H \times G), i = 1, 2, \dots, \quad (3)$$

$$z_r \sim \bar{\pi}_{z_{r-1}}, r = 1, 2, \dots, \quad (4)$$

$$d_r \sim g(\vartheta_{z_r}), r = 1, 2, \dots, \quad (5)$$

$$x_{t_r^1:t_r^2} = z_r \quad (6)$$

$$y_{t_r^1:t_r^2} \sim h(\varphi_{z_r}) \quad (7)$$

where GEM denotes a stick-breaking process, α and γ are concentration parameters. $h(\cdot)$ and $g(\cdot)$ are observation and duration distributions, respectively, while H and G denote the prior distribution of the corresponding parameters (i.e., φ and ϑ). $\bar{\pi}_i \triangleq \frac{\pi_{ij}}{1 - \pi_{ii}} (1 - \delta_{ij})$ is adopted to eliminate self-transition in the super-state sequence, and π_{ij} is the transition probability from states (driving regimes) i to j . $x_{t_r^1:t_r^2}$ and $y_{t_r^1:t_r^2}$ represent the label and observation segments associated with the super-state z_r , where t_r^1 and t_r^2 denote the start and end time indices of the corresponding time segment t_r .

3.3. Multi-regime car-following model and its calibration

3.3.1. Intelligent driver model and probabilistic intelligent driver model

The IDM is a representative desired-acceleration car-following model (Treiber et al., 2000), characterized by a continuous nonlinear function that maps spacing (s), speed (v), and speed difference (Δv) to acceleration a_{IDM} at each time step. It incorporates speed and spacing, providing a unified functional form to describe traffic dynamics under both free-flow and congested conditions. The model is formulated as follows:

$$a_{\text{IDM}}^{(t)} = \mathcal{G}(s_n^{(t)}, v_n^{(t)}, \Delta v_n^{(t)}) = a_m \left[1 - \left(\frac{v_n^{(t)}}{v_m} \right)^\delta - \left(\frac{s^*(v_n^{(t)}, \Delta v_n^{(t)})}{s_n^{(t)}} \right)^2 \right] \quad (8)$$

$$s^*(v_n^{(t)}, \Delta v_n^{(t)}) = s_0 + s_1 \sqrt{\frac{v_n^{(t)}}{v_m}} + v_n^{(t)} T + \frac{v_n^{(t)} \Delta v_n^{(t)}}{2\sqrt{a_m b_m}} \quad (9)$$

where v_m , s_0 , T , a_m , b_m , and δ are model parameters with specific physical meanings. v_m represents the maximum desired speed, commonly referred to as the free-flow speed; s_0 denotes the minimum safe distance (gap) maintained from the leading vehicle; T

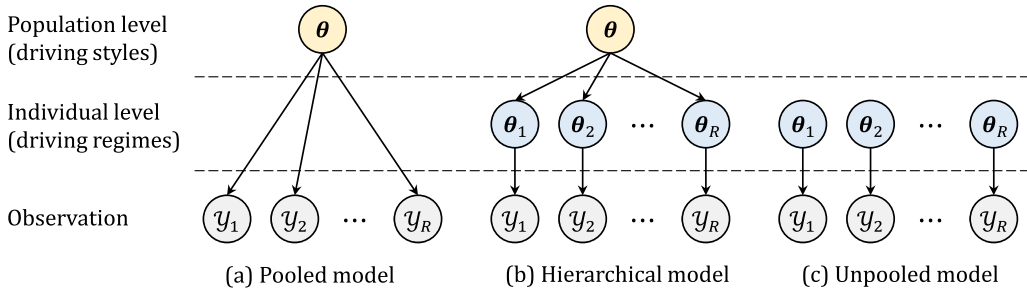


Fig. 3. Multi-regime pooled, hierarchical, and unpooled models.

represents the desired safe time headway, which is the time gap between the following and leading vehicles; a_m and b_m are the maximum desired acceleration and comfort deceleration, respectively. δ denotes the acceleration exponent, which is set to $\delta = 4$ by default. The vehicle deceleration is controlled by the desired minimum gap s^* in IDM, where s_1 is set to 0 to derive a model with interpretable and easily measurable parameters. $\mathcal{G}(\cdot)$ represents the IDM function.

However, the imperfections and irregularities in driving behavior lead to instability and randomness in driver behavior, as mentioned by Treiber and Kesting (2012). To address this issue, a random term with a standard deviation σ_ϵ is incorporated into the acceleration $a_{\text{IDM}}^{(t)}$ to account for imperfect driving behaviors. For brevity, the parameter vector of IDM is defined as $\theta = [v_m, s_0, T, a_m, b_m] \in \mathbb{R}^5$, and the inputs are denoted as $\mathbf{y}^{(t)} = [s^{(t)}, v^{(t)}, \Delta v^{(t)}]$, in which the following vehicle index n is ignored for the sake of simplicity. Consequently, a stochastic version of IDM (i.e., probabilistic IDM) is formulated as follows:

$$a^{(t)} = a_{\text{IDM}}^{(t)} + \epsilon_t, \epsilon_t \sim \mathcal{N}^{i.i.d.}(0, \sigma_\epsilon^2) \quad (10)$$

where ϵ_t denotes the random term (noise) in acceleration, which is independent and identically distributed (i.i.d.) with a Gaussian distribution. Consequently, the formulation can be rewritten as $a^{(t)} | \mathbf{y}^{(t)}, \theta \sim \mathcal{N}^{i.i.d.}(a_{\text{IDM}}^{(t)}, \sigma_\epsilon^2)$, in which a_{IDM} and σ_ϵ^2 denote the mean and variance of the acceleration, respectively. Here, an integration scheme proposed by Treiber and Kesting (2013) is adopted to update the vehicle speed and position with a time step Δt :

$$v^{(t+\Delta t)} = v^{(t)} + a^{(t)} \cdot \Delta t \quad (11)$$

$$x^{(t+\Delta t)} = x^{(t)} + \frac{v^{(t+\Delta t)} + v^{(t)}}{2} \cdot \Delta t \quad (12)$$

3.3.2. Multi-regime pooled, hierarchical, and unpooled B-IDM

Zhang and Sun (2024) proposed pooled, hierarchical, and unpooled car-following models, founded on the assumption that human drivers maintain consistent driving styles within a defined driving condition, yet demonstrate distinct driving styles across individuals (drivers). Under this assumption, the hierarchical model is calibrated using a Bayesian approach to capture heterogeneous car-following behaviors among diverse human drivers. However, human driving styles may dynamically change across different driving regimes during car-following (Zou et al., 2022; Rao et al., 2023; Tang et al., 2025). Therefore, this study extends the hierarchical modeling framework by incorporating variations in driving styles across different driving regimes and develops multi-regime pooled, hierarchical, and unpooled car-following models accordingly. The primary distinction between these two modeling frameworks lies in their conceptual foundations and hierarchical interpretation. The work by Zhang and Sun (2024) is grounded in driver diversity, focusing on inter-driver heterogeneity by assigning individualized parameter sets to represent each human driver. In contrast, our framework accounts for both inter-driver and intra-driver heterogeneity: the population level represents variation across distinct driving styles (e.g., neutral, timid, and aggressive), whereas the individual level captures behavioral heterogeneity across different driving regimes within drivers of the same driving style. Moreover, inter-driver heterogeneity is modeled at the individual level in Zhang and Sun (2024), whereas our framework represents it at the driving style level.

Fig. 3 illustrates the differences among the pooled, hierarchical, and unpooled models. In the pooled model, all drivers sharing the same driving style are assumed to exhibit identical driving behaviors across driving regimes ($\forall r \in \{1, 2, \dots, R\}$), as characterized by a common parameter set θ . In the hierarchical model, it is assumed that drivers with the same driving style exhibit identical driving behaviors under the same driving regime, while their behaviors vary across different driving regimes. In this assumption, each driving regime is characterized by a distinct parameter set θ_r , yet all are influenced by the universal population level (driving style). Generally, the hierarchical model defines a population-level prior that governs the parameter set θ_r for each specific driving regime. As a result, the parameter sets θ_r for driving regimes are not estimated independently, allowing for a more precise representation of behavioral variation across diverse driving regimes. In contrast, the unpooled model assigns a completely independent parameter set θ_r to each driving regime.

For clarity, we present the formulations of the three models, along with their corresponding probabilistic graphical models, as illustrated in Fig. 4. For the pooled B-IDM, the formulations are represented as follows:

$$\ln(\theta) \sim \mathcal{N}(\mu_0, \Sigma_0) \in \mathbb{R}^5 \quad (13)$$

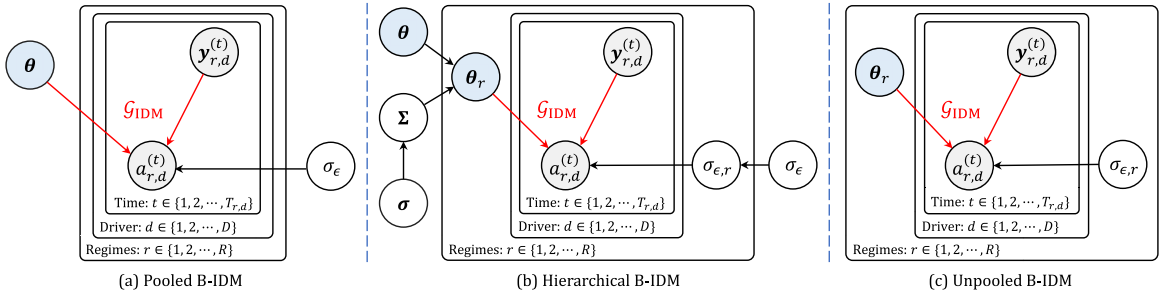


Fig. 4. Probabilistic graphical models for multi-regime pooled, hierarchical, and unpooled B-IDM. Where θ denotes the shared parameter set associated with a driving style and θ_r represents the parameter for each driving regime.

$$\ln(\sigma_\epsilon) \sim \mathcal{N}(\mu_\sigma, \sigma_0^2) \in \mathbb{R} \quad (14)$$

for regime $r = 1, \dots, R$:

for driver $d = 1, \dots, D$:

for time $t = 1, \dots, T_{r,d}$:

$$a_{r,d}^{(t)} | y_{r,d}^{(t)}, \theta \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(a_{\text{IDM}}^{(t)}, \sigma_\epsilon^2) \in \mathbb{R} \quad (15)$$

where μ_0 , Σ_0 , μ_σ , and σ_0 are treated as manually specified hyperparameters. Specifically, μ_0 is the prior mean of IDM parameters and is defined as $\mu_0 = \ln(\theta_{\text{rec}})$. Here, to avoid unrealistic negative sampled values during Markov chain sampling (Abodo et al., 2019), both θ and σ_ϵ are assumed to follow a log-normal distribution.

For the unpooled B-IDM, the goal is to estimate a specific parameter set θ_r for each driving regime. For each driving regime r ($r = 1, \dots, R$), the formulations are provided as follows:

$$\ln(\theta_r) \sim \mathcal{N}(\mu_0, \Sigma_0) \in \mathbb{R}^5 \quad (16)$$

$$\ln(\sigma_{\epsilon,r}) \sim \mathcal{N}(\mu_\sigma, \sigma_0^2) \in \mathbb{R} \quad (17)$$

for driver $d = 1, \dots, D$:

for time $t = 1, \dots, T_{r,d}$:

$$a_{r,d}^{(t)} | y_{r,d}^{(t)}, \theta_r \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(a_{\text{IDM}}^{(t)}, \sigma_{\epsilon,r}^2) \in \mathbb{R} \quad (18)$$

Note that the primary distinction between pooled and unpooled models lies in their data requirements. The pooled model requires the entire dataset to estimate a common parameter set θ , whereas the unpooled model estimates a driving regime-specific parameter set θ_r using only data associated with driving regime r .

The pooled model assumes a single common parameter set shared across all drivers and driving regimes, capturing only the general car-following behavior at the population level (i.e., driving style). This limits its ability to represent intra-heterogeneity across different driving regimes. In contrast, the unpooled model estimates independent parameter sets θ_r for each driving regime, allowing it to characterize car-following behavior under distinct driving regimes. However, it does not adequately model the general driving style that shapes driving behavior across different driving regimes. Therefore, a hierarchical B-IDM is developed to explicitly represent general driving behavior at the population level, while simultaneously accounting for behavioral heterogeneity across different driving regimes at the individual level. Note that the hierarchical model assigns a parameter set not to each individual human driver, but rather to each driving regime, i.e., θ_r . The formulations are summarized as follows:

$$\sigma \sim \text{Exp}(\lambda) \in \mathbb{R}^5 \quad (19)$$

$$\Sigma \sim \text{LKJCholeskyCov}(\eta, \sigma) \in \mathbb{R}^{5 \times 5} \quad (20)$$

$$\ln(\theta) \sim \mathcal{N}(\mu_0, \Sigma_0) \in \mathbb{R}^5 \quad (21)$$

$$\sigma_\epsilon \sim \text{Exp}(\lambda_\epsilon) \in \mathbb{R} \quad (22)$$

for regime $r = 1, \dots, R$:

$$\ln(\theta_r) \sim \mathcal{N}(\ln(\theta), \Sigma) \in \mathbb{R}^5 \quad (23)$$

$$\ln(\sigma_{\epsilon,r}) \sim \mathcal{N}(\ln(\sigma_\epsilon), \sigma_\sigma^2) \in \mathbb{R} \quad (24)$$

for driver $d = 1, \dots, D$:

for time $t = 1, \dots, T_{r,d}$:

$$a_{r,d}^{(t)} | y_{r,d}^{(t)}, \theta_r \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(a_{\text{IDM}}^{(t)}, \sigma_{\epsilon,r}^2) \in \mathbb{R} \quad (25)$$

where LKJ refers to the LKJ distribution, and LKJCholeskyCov denotes a distribution over Cholesky decomposed covariance matrices, such that the underlying correlation matrices follow an LKJ distribution and the standard deviations follow an exponential distribution (Salvatier et al., 2016). $\text{Exp}(\lambda)$ denotes the exponential distribution with rate λ .

3.3.3. Bayesian calibration of B-IDM

The calibration process of the car-following model aims to estimate the parameter set θ , which defines the optimal mapping of model inputs to a defined measure of performance, such as acceleration, speed, and spacing (Sharma et al., 2019). Previous studies have proposed various approaches to calibrate the parameters of car-following models, with heuristic algorithms and maximum likelihood methods being among the most commonly adopted techniques (Treiber and Kesting, 2013; Punzo et al., 2021). In this study, we adopt a Bayesian approach developed by Zhang and Sun (2024) to calibrate the model parameters of multi-regime pooled, hierarchical, and unpooled B-IDM. The approach is inherently a sampling-based calibration method, and the sampling procedures are described in detail in Section 3.3.2. The likelihood function critically impacts parameter calibration by linking observed data to parameter inference, and its formulation is closely related to the chosen measure of performance.

For acceleration a as the measure of performance, the likelihood at each time step t can be formulated as follows:

$$\hat{a}^{(t)} \sim \mathcal{N}(a_{\text{IDM}}^{(t)}, \sigma_e^2) \quad (26)$$

where $\hat{a}$ denotes the observed acceleration data. In this study, the speed v is selected as the measure of performance for calibrating model parameters. According to the updated scheme in Eq. (11), the speed at time $t + \Delta t$ is computed as $v^{(t+\Delta t)} = v^{(t)} + a_{\text{IDM}}^{(t)} \Delta t + \epsilon_t \Delta t$, $\epsilon_t \Delta t \sim \mathcal{N}(0, (\sigma_e \Delta t)^2)$. Consequently, the likelihood of speed can be formulated as follows:

$$\hat{v}^{(t+\Delta t)} \sim \mathcal{N}(v_{\text{IDM}}^{(t+\Delta t)}, (\sigma_e \Delta t)^2) \quad (27)$$

where $\hat{v}^{(t+\Delta t)}$ represents the observed speed data, and $v_{\text{IDM}}^{(t+\Delta t)} = v^{(t)} + a_{\text{IDM}}^{(t)} \Delta t$.

4. Data preparation

4.1. Waymo open dataset

4.1.1. Data description

The Waymo open dataset comprises high-resolution sensor data collected on a large scale from Waymo's autonomous vehicles (AVs), covering diverse locations, such as San Francisco, Seattle, and Los Angeles (Sun et al., 2020; Ettinger et al., 2021). The dataset covers a wide array of driving conditions, including various road types (i.e., freeways, urban streets), weather conditions (e.g., rainy, sunny), and times of day (e.g., daytime, nighttime). The dataset comprises perception and motion datasets, but only the motion dataset is utilized in this study. The motion dataset, stored in TFRecord format, contains protocol buffer data and is available for download from the Waymo open dataset website (<https://waymo.com/open/data/download/>). The dataset consists of 103,354 segments (scenarios), each spanning 20 s of object tracking data recorded at a frequency of 10 Hz, along with map data for the corresponding geographic area. In addition, the dataset encompasses a total of 574 h and covers a distance of 1750 km. The sensor data were gathered using 1 mid-range LiDAR and 4 short-range LiDARs, while the object data were provided through 3D ground truth bounding boxes for each detected object (e.g., vehicles, pedestrians, and cyclists). The motion dataset offers detailed information, including scenario IDs, timestamps, object tracks (comprising object IDs, object types, and motion states such as position, velocity, heading, etc.), map features (e.g., lane centers, lane boundaries, etc.), among other relevant attributes. In addition, the motion dataset is partitioned into training, validation, and test sets, with allocations of 70%, 15%, and 15%, respectively, to serve distinct purposes. Given that only the training dataset contains the full 20 s of information, this study focuses on the training set, which consists of 70,541 segments.

4.1.2. Data processing

According to the research purpose of car-following behavior analysis, a data processing scheme proposed by Hu et al. (2022) is utilized to improve data quality. The procedures are described as follows:

- Transforming the hierarchical data structure (i.e., TFRecord format) into a tabular data structure (e.g., CSV format);
- Extracting the motion trajectories of AV and its surrounding human-driven vehicles (HVs), excluding cyclists and pedestrians;
- Generating the videos according to the extracted vehicle motion trajectories and the corresponding map data;
- Manually identifying appropriate car-following vehicle pairs from videos by verifying that both the leading and following vehicles are traveling in the same lane;
- Enhancing data quality through outlier removal via an optimization-based approach and smoothing using a wavelet filter.

4.2. Lyft Level-5 dataset

The Lyft Level-5 dataset is a large-scale autonomous driving dataset (Houston et al., 2021), collected over 4 months by 20 self-driving cars along a fixed route in Palo Alto, California. It comprises 170,000 scenes, each lasting 25 s, resulting in 1118 h of recorded logs. The sensors for perception in self-driving vehicles include 7 cameras, 3 LiDARs, and 5 radars, with a sampling rate of 10 Hz. In each scene, agent track IDs are renumbered starting from 1. The dataset for each scene encompasses detailed information, including the position, yaw angle, size, and speed of each detected vehicle, cyclist, and pedestrian, as well as the position and orientation of the autonomous vehicle (AV).

For car-following behavior analysis, Li et al. (2023) processed and refined the Lyft level-5 dataset to conduct a comparative study of driving behaviors exhibited by HVs and AVs. The processed dataset is available at <https://github.com/RomainLITUD/Car-Following-Dataset-HV-vs-AV>. The shared dataset comprises 38,683 HV-following-HV (HV–HV) events and 26,394 HV-following-AV (HV–AV) events. Leveraging this dataset, this study employed an optimization-based approach for outlier removal and applied wavelet filtering to improve data smoothness, thereby enhancing overall data quality.

4.3. Car-following event extraction

The qualified data is fundamental to car-following behavior modeling. Previous studies have proposed various criteria for extracting car-following pairs, primarily concerning road type, time headway, spacing, speed of the following vehicle, and duration. Considering the timespan of each segment and sensor detection range, the qualified car-following events are extracted according to the following criteria (Hammit et al., 2018; Hu et al., 2022; Wen et al., 2022):

- The leading and following vehicles must drive on a straight freeway segment, as urban street scenarios are excluded from this study. This criterion is only for the Waymo dataset.
- The maximum spacing between the leading and following vehicles during car-following should be less than 85 m;
- The speed of the following vehicle must exceed 3 m/s to mitigate the influences of traffic jam conditions;
- The duration of a car-following event should be at least 15 s for the Waymo dataset, and at least 20 s for the Lyft dataset
- The lengths of both the leading and following vehicles must be greater than 3 m and less than 6 m.

Finally, 459 HV–HV events and 71 HV–AV events are extracted from the Waymo open motion dataset. Moreover, due to the large size of the Lyft level-5 dataset, this study randomly selects and reassigns identifiers to 1000 HV–HV events and 1000 HV–AV events from the extracted car-following data, in accordance with the specified criteria. Note that only the car-following data for HV–HV and HV–AV modes are applied in this study to model human car-following behaviors.

5. Experiments and results

5.1. Classification of driving styles

The classification of driving styles is founded on the assumption that driving behavior is observable, measurable, and categorizable into a limited number of distinct types (e.g., neutral, aggressive, timid, etc.), and that driving styles exhibit temporal stability. Previous studies have highlighted that average time headway (Ave. Thw), maximum acceleration (Max. Acc), and maximum deceleration (Max. Dec) are three critical parameters in characterizing car-following behavior (Chen et al., 2020). In addition, Chen et al. (2020) compared 6 clustering methods in classifying driving styles and demonstrated that the K-means clustering method achieves the best performance. Therefore, considering average time headway, maximum acceleration, and maximum deceleration, this study applies the K-means clustering method to classify driving styles in the context of car-following behavior. The clustering results are provided in Table 1, with a silhouette coefficient of 0.322 for the Waymo dataset and 0.337 for the Lyft dataset. In conjunction with the classification diagrams presented in Figs. 5 and 6, the differences among the three driving styles are distinctly observable.

The aggressive driving style is characterized by an approximately minimal time headway, as well as the highest acceleration and deceleration, which aligns with the conventional understanding of aggressive driving behavior. In general, aggressive drivers tend to travel at higher speeds and maintain shorter following distances from the leading vehicle. In emergencies, they are more likely to apply stronger deceleration, followed by a spontaneous increase in acceleration to reach the desired speed. For the neutral driving style, drivers tend to select a slightly longer time headway compared to aggressive drivers. Their acceleration and deceleration behaviors are more moderate, aiming to balance safety, efficiency, and comfort. In contrast, timid drivers exhibit the longest time headway, along with low acceleration and deceleration, indicating a preference for maintaining safety by keeping a greater distance from the leading vehicle. It is noted that the vehicle deceleration exhibited by timid drivers is slightly higher than that of neutral drivers. This may be due to their heightened tension and lack of confidence, leading to cautiousness in maintaining safety during braking. In addition, the driving characteristics associated with the three driving styles may vary across different driving scenarios and are not necessarily consistent. For instance, in the Waymo dataset (i.e., freeway scenario), the aggressive driving style is associated with the shortest time headway. In contrast, within the Lyft dataset (i.e., urban road scenarios), the neutral driving style demonstrates a slightly longer time headway compared to the aggressive style. The average time headways for the neutral and aggressive driving styles are 1.895 s and 1.780 s, respectively, in the Waymo dataset, and 1.689 s and 1.789 s, respectively, in the Lyft dataset.

Table 1
Clustering center of K-means.

Clustering center	Waymo dataset			Lyft dataset		
	Neutral	Timid	Aggressive	Neutral	Timid	Aggressive
Ave. Thw (s)	1.895	3.346	1.780	1.689	3.467	1.789
Max. Acc (m/s^2)	0.948	0.756	1.411	0.654	0.880	1.471
Max. Dec (m/s^2)	-0.749	-0.774	-2.228	-0.589	-0.658	-1.377
Proportion (%) of HV-HV	264 (57.52%)	100 (21.79%)	95 (20.70%)	552 (55.20%)	180 (18.00%)	268 (26.80%)
Proportion (%) of HV-AV	37 (52.11%)	18 (25.35%)	16 (22.54%)	595 (59.50%)	132 (13.20%)	273 (27.30%)

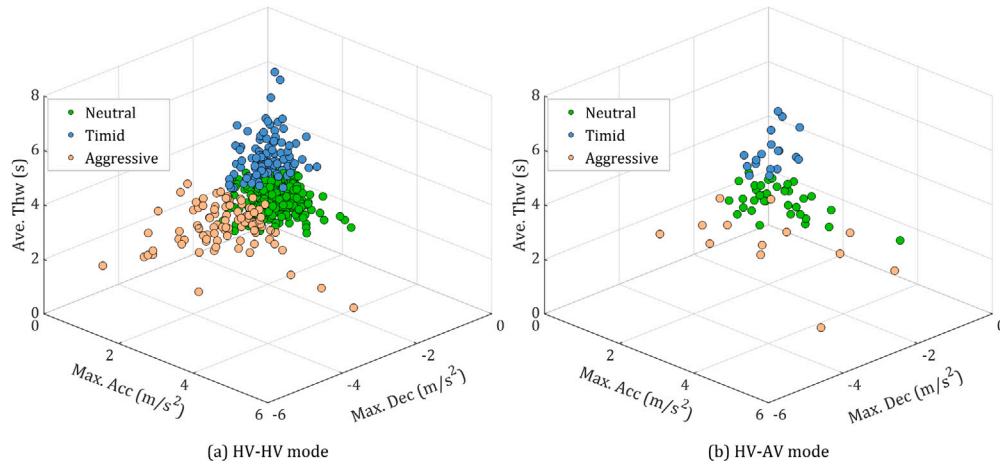


Fig. 5. Classification diagrams of driving styles for the Waymo dataset.

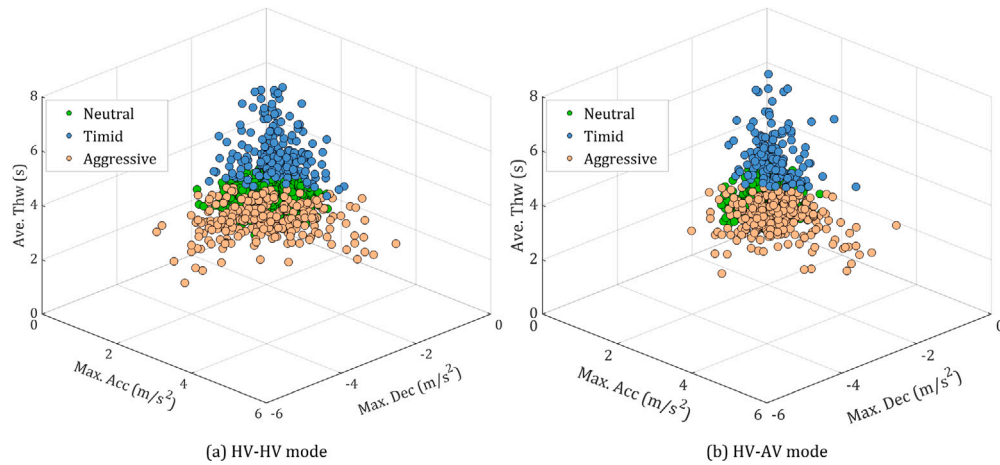


Fig. 6. Classification diagrams of driving styles for the Lyft dataset.

5.2. Segmentation of driving regimes

This study employs a Gibbs sampler for inference in the HDP-HSMM to segment driving regimes during car-following. The Poisson and Gaussian distributions are chosen as the observation and state duration distributions (Zou et al., 2022), respectively. Among the hyperparameters, the concentration parameters α and γ are set to 4. In general, car-following regimes are classified into 4 types (Sharma et al., 2018; Hu et al., 2023), which are following the leader at a constant speed (F), accelerating behind a leader (A), decelerating behind a leader (D), and standing behind a leader (S). Given that the extracted car-following events do not involve standing states, this study limits the maximum number of states to 3.

To illustrate the behavioral differences across various driving regimes, four driving behavior parameters, acceleration, spacing, speed, and speed difference, are adopted as metrics. To avoid the issue of average acceleration approaching zero in longer episodes, we instead use absolute acceleration to reflect overall driving intensity. Table 2 summarizes the means of driving behavior

Table 2

Summary of average values for driving behavior parameters.

Data	Mode	Regime	Neutral				Timid				Aggressive			
			a (m/s ²)	v (m/s)	s (m)	Δv (m/s)	a (m/s ²)	v (m/s)	s (m)	Δv (m/s)	a (m/s ²)	v (m/s)	s (m)	Δv (m/s)
Waymo	HV–HV	LS	0.37	10.37	13.58	−0.23	0.37	7.79	20.14	−0.19	0.69	10.76	13.89	0.03
		HS	0.38	13.17	20.01	0.02	0.30	10.40	29.86	0.14	0.59	16.21	23.22	0.59
		MS	0.39	12.22	16.54	−0.03	0.35	9.05	24.65	0.10	0.79	13.63	17.25	0.23
	HV–AV	LS	0.44	12.96	16.77	−0.27	0.41	8.58	23.94	−0.12	0.54	13.98	17.06	0.05
		HS	0.41	16.56	27.12	0.17	0.34	11.24	36.08	0.18	0.71	19.96	27.43	0.91
		MS	0.49	13.84	21.14	−0.03	0.35	12.05	36.63	−0.14	1.11	16.07	20.87	0.45
Lyft	HV–HV	LS	0.35	12.97	17.55	−0.25	0.43	11.79	33.53	−0.60	0.62	11.36	15.41	−0.10
		HS	0.25	15.53	24.93	0.01	0.25	14.48	47.92	−0.32	0.39	14.92	23.39	0.19
		MS	0.28	14.45	20.92	−0.07	0.29	13.77	42.18	−0.42	0.50	13.69	18.87	0.10
	HV–AV	LS	0.30	11.32	11.99	−0.11	0.44	9.22	23.81	−0.21	0.61	8.68	9.19	−0.15
		HS	0.21	12.93	17.29	0.25	0.24	11.76	40.36	0.62	0.36	12.03	18.45	0.46
		MS	0.23	12.55	14.71	0.02	0.33	11.23	32.74	0.42	0.50	11.31	14.36	0.15

parameters, with the corresponding boxplots illustrated in Fig. 7 for the Waymo dataset and Fig. A.12 for the Lyft dataset. In contrast to previous studies (Sharma et al., 2018; Hu et al., 2023), this study incorporates spacing, speed, and speed difference to segment driving regimes, with obvious differences observed in spacing and speed. Consequently, the driving regimes are categorized based on their average speed, with regime #2 exhibiting the highest speed and labeled as high-speed (HS), regime #1 showing the lowest speed and labeled as low-speed (LS), and regime #3 falling in between and labeled as moderate-speed (MS).

From Table 2, significant differences in driving behavior are observed across various driving regimes during car-following, as well as among different driving styles and car-following modes. In general, Regime #1 (LS) is characterized by the lowest speed, spacing, and speed difference, along with moderate acceleration/deceleration. In contrast, Regime #2 (HS) exhibits the highest values of speed, spacing, and speed difference, but is accompanied by the smallest acceleration/deceleration. Regime #3 (MS), by comparison, is defined by moderate speed and spacing, along with a moderate speed difference. For example, in the Waymo dataset, the average speeds for the LS, HS, and MS regimes under the HV–HV mode and neutral driving style are 10.37 m/s, 13.17 m/s, and 12.22 m/s, respectively. In terms of driving style, the aggressive driving style exhibits the highest acceleration/deceleration, the highest speed, and the greatest speed difference compared to the neutral and timid driving styles. Taking the Waymo dataset as an example, in the HV–HV mode, the average acceleration/deceleration values for the aggressive driving style in the LS, HS, and MS regimes are 0.69 m/s², 0.59 m/s², and 0.79 m/s², respectively, each of which exceeds the corresponding values observed for the neutral and timid styles. An interesting finding is that the speed and spacing associated with different car-following modes vary across driving scenarios. In the freeway scenario (i.e., the Waymo dataset), the HV–AV mode results in higher speed and greater spacing compared to the HV–HV mode. In contrast, the HV–HV mode exhibits higher speed and spacing than the HV–AV mode in the urban road scenario (i.e., the Lyft dataset). For instance, in the freeway scenario, the average spacings in the LS, HS, and MS regimes for neutral drivers under the HV–HV mode are 16.77 m, 27.12 m, and 21.14 m, respectively, each of which is greater than the corresponding value observed under the HV–HV mode. These findings suggest that it is necessary to distinguish between driving styles and driving regimes when modeling heterogeneous human car-following behaviors.

5.3. Calibration results and analysis

In this study, PyMC is adopted to calibrate the model parameters. Specifically, the No-U-Turn Sampler (NUTS) is employed to generate posterior samples, with 5000 burn-in steps to ensure the convergence of the Markov chains, and 2500 posterior samples are drawn to estimate the posterior distributions. According to the study by Punzo et al. (2014), the priors for the IDM parameter are set as $\theta_{\text{rec}} = [33.3, 2.0, 1.6, 1.5, 1.67]$ for the Waymo dataset, and $\theta_{\text{rec}} = [25.0, 2.0, 1.6, 1.5, 1.67]$ for the Lyft dataset, with the population prior set as $\mu_0 = \ln(\theta_{\text{rec}})$. Moreover, the other hyperparameters are set following the recommendations of Zhang and Sun (2024). In this study, we set $\lambda = 100$ for the exponential distribution, along with $\eta = 2$ for the LKJCholeskyCov distribution. For the hyperparameter σ_e , we specify $\lambda_e = 1 \times 10^4$ for the exponential distribution to scale the error variance. At the individual level (i.e., driving regime), the prior of $\sigma_{e,r}$ is set with $\sigma_\sigma = 0.05$. Note that the parameter λ controls the variance of the IDM parameters at the population level (i.e., driving style). A large value of λ implies a small variance, resulting in the hierarchical model behaving more like a pooled model and vice versa.

5.3.1. Calibrated results of IDM parameters

According to the Bayesian calibration procedure, a set of posterior samples was obtained. To compare clearly, the posterior expectations of each parameter for the Waymo dataset are summarized in Table 3, and the results for the Lyft data are presented in Table B.6. It is evident that all the calibrated parameters fall within the recommended ranges, as suggested by previous studies (Kesting and Treiber, 2008; Kim and Mahmassani, 2011). The results demonstrate the reliability of the Bayesian approach in calibrating model parameters. According to the calibrated results, some interesting findings are discussed as follows:

(1) It is evident that the desired safe time headway T for the aggressive driving style is shorter than that of the neutral and timid styles, while the comfort vehicle deceleration b_m is larger than that of the neutral and timid styles. These results are consistent with

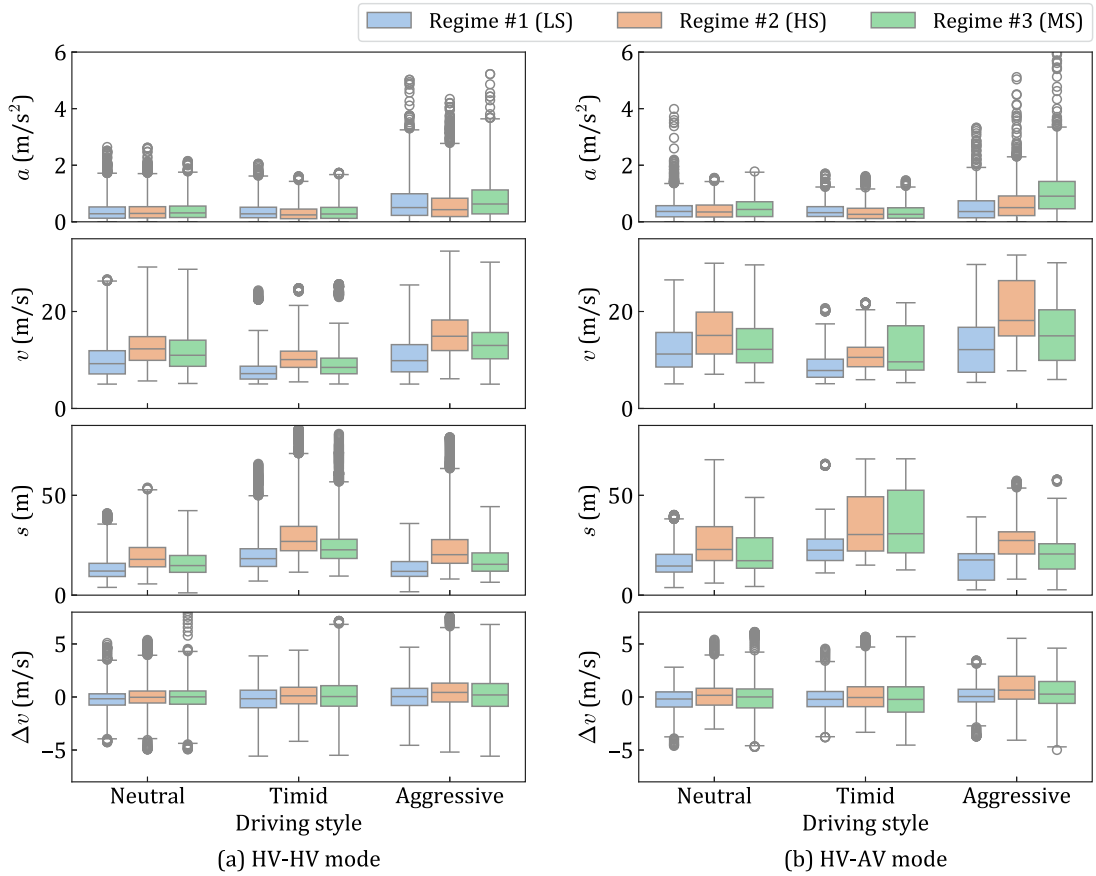


Fig. 7. Boxplots of driving parameters for the Waymo dataset.

the classification of driving styles, indicating that aggressive drivers are more likely to engage in more assertive driving behaviors, such as maintaining a shorter time headway and applying higher deceleration to ensure safety. In addition, a distinct difference between neutral and timid driving styles is evident in the minimum safe distance s_0 , with timid drivers maintaining a larger distance than neutral drivers. Furthermore, among different driving regimes, the HS regime exhibits the shortest time headway, followed by the MS and LS regimes in ascending order. For instance, in the Waymo dataset, the safe time headways of the hierarchical B-IDM under the HV-HV mode and neutral driving style are 0.812 s, 0.469 s, and 0.690 s for the LS, HS, and MS regimes, respectively.

(2) The parameters of the pooled B-IDM represent general driving behavior at the population level. Specifically, the pooled model estimates a single set of parameters to describe all driving behaviors within the dataset that includes diverse driving regimes. However, human driving behavior varies across different regimes. The key limitation of the pooled model is that it cannot explain the heterogeneous driving behavior across different driving regimes. As a result, the pooled model is assigned a significant variance σ_e^2 to account for uncertainty in driving behaviors. In contrast, the unpooled model exhibits a smaller residual variance σ_e^2 , since it creates separable models for each specific driving regime, though at the risk of overfitting. In the hierarchical model, the variance σ_e^2 at the population level (i.e., across driving styles) is typically smaller than that at the individual level (i.e., across driving regimes). This is because the population level represents relatively stable, long-term behavioral tendencies that differ only slightly across driving styles. In contrast, the individual level captures short-term, context-dependent behavioral fluctuations within a given driving style.

(3) The calibrated parameters of the hierarchical B-IDM closely resemble those of the unpooled model at the individual level (driving regime), while aligning with the pooled model at the population level (driving style). These findings suggest consistency across the proposed models with different hierarchies. For intuition, the kernel density estimations of calibrated IDM parameters for the hierarchical model, derived from the Waymo dataset, are illustrated in Fig. 8, while those for the Lyft dataset are presented in Fig. B.13. The solid curve represents the HV-HV mode, while the dotted curve indicates the HV-AV mode. The results demonstrate that model parameters exhibit significant variations across different car-following modes and driving regimes. This observation aligns with the findings of previous studies (Punzo and Montanino, 2020; Kang et al., 2023). Moreover, the findings support the necessity of dividing driving regimes when modeling heterogeneous car-following behaviors. The hierarchical models are capable of simultaneously identifying style-specific parameters across different driving regimes.

Table 3

Posterior expectations of parameters for pooled, hierarchical, and unpooled B-IDM in the Waymo dataset.

Style	Model	Regime	HV–HV mode		HV–AV mode	
			$\theta = [v_m, s_0, T, a_m, b_m]$	σ_ϵ	$\theta = [v_m, s_0, T, a_m, b_m]$	σ_ϵ
Neutral	PB-IDM	Pop	[26.260, 2.266, 0.555, 0.651, 1.757]	0.266	[18.325, 1.942, 0.861, 0.567, 1.905]	0.293
		Pop	[27.507, 1.656, 0.757, 0.616, 2.286]	0.114	[20.092, 2.021, 0.860, 0.560, 1.769]	0.121
	HB-IDM	LS	[27.300, 1.634, 0.812, 0.469, 2.430]	0.292	[16.785, 2.038, 0.720, 0.225, 1.826]	0.342
		HS	[26.801, 1.615, 0.469, 0.463, 2.561]	0.297	[19.165, 2.014, 0.696, 0.716, 1.893]	0.330
		MS	[27.563, 1.650, 0.690, 0.602, 2.282]	0.289	[20.087, 2.021, 0.888, 0.526, 1.660]	0.326
		LS	[24.062, 2.010, 0.698, 0.677, 1.731]	0.200	[15.112, 1.938, 0.847, 0.509, 1.949]	0.239
	UB-IDM	HS	[25.468, 1.905, 0.396, 0.666, 1.702]	0.203	[19.660, 1.942, 0.868, 0.708, 1.894]	0.211
		MS	[26.284, 1.890, 0.603, 0.704, 1.722]	0.184	[16.202, 1.945, 0.934, 0.645, 1.750]	0.204
Timid	PB-IDM	Pop	[19.431, 2.312, 1.567, 0.651, 1.743]	0.272	[21.425, 2.023, 1.740, 0.652, 1.718]	0.310
		Pop	[26.696, 6.485, 0.802, 0.501, 1.625]	0.114	[24.342, 2.180, 1.861, 0.491, 2.095]	0.123
	HB-IDM	LS	[26.812, 5.350, 0.982, 0.475, 1.616]	0.321	[25.297, 2.288, 2.268, 0.300, 2.071]	0.348
		HS	[26.870, 3.779, 0.817, 0.336, 1.714]	0.290	[23.915, 1.975, 1.640, 0.581, 2.223]	0.335
		MS	[26.807, 4.727, 1.039, 0.347, 1.536]	0.280	[20.927, 2.077, 1.881, 0.297, 2.138]	0.351
		LS	[17.947, 2.470, 1.616, 0.696, 1.691]	0.212	[20.752, 2.036, 1.821, 0.674, 1.695]	0.243
	UB-IDM	HS	[18.314, 1.725, 1.497, 0.681, 1.738]	0.207	[22.680, 1.997, 1.624, 0.713, 1.697]	0.213
		MS	[15.988, 2.152, 1.578, 0.668, 1.717]	0.199	[20.640, 2.006, 1.680, 0.665, 1.696]	0.247
Aggressive	PB-IDM	Pop	[29.569, 2.671, 0.581, 0.696, 3.589]	0.377	[29.126, 1.907, 0.392, 0.628, 3.496]	0.577
		Pop	[27.308, 4.829, 0.293, 0.746, 3.385]	0.141	[31.184, 2.022, 0.670, 0.282, 3.210]	0.165
	HB-IDM	LS	[28.869, 5.886, 0.482, 0.416, 4.525]	0.446	[30.253, 2.120, 1.108, 0.248, 2.780]	0.558
		HS	[18.936, 2.907, 0.351, 0.773, 3.812]	0.451	[33.164, 1.830, 0.243, 0.430, 3.799]	0.556
		MS	[34.050, 5.830, 0.440, 0.727, 3.736]	0.467	[31.313, 1.919, 0.578, 0.274, 3.546]	0.882
		LS	[26.151, 2.796, 0.645, 0.683, 3.980]	0.285	[23.097, 1.894, 0.483, 0.615, 1.824]	0.388
	UB-IDM	HS	[19.498, 1.707, 0.434, 0.741, 2.595]	0.263	[31.026, 1.914, 0.327, 0.675, 4.049]	0.369
		MS	[33.542, 2.296, 0.567, 0.717, 2.863]	0.282	[27.485, 1.881, 0.439, 0.670, 2.712]	0.486

Note: PB-IDM, HB-IDM, and UB-IDM denote the pooled, hierarchical, and unpooled B-IDM, and Pop denotes the population level. The recommendation value: $\theta_{\text{rec}} = [33.3, 2.0, 1.6, 1.5, 1.67]$.

5.3.2. Correlation matrices of IDM parameters

To investigate the IDM parameter correlations, the correlation matrices of parameters for the hierarchical B-IDM are illustrated in Fig. 9, using the neutral driving style from the Waymo dataset as an example. In general, the correlations between HV–HV and HV–AV modes are similar. Strong positive correlations are observed in the pairs of (v_m, T) , (v_m, b_m) , and (T, b_m) . In contrast, significantly negative correlations are found in the pairs of (v_m, s_0) , (s_0, T) , and (s_0, b_m) . These correlations are fundamentally consistent with those reported in previous studies (Treiber and Kesting, 2013), where the calibrations were based on the likelihood rather than the posterior distribution. However, the correlation of the pair (a_m, b_m) differs across the two car-following modes, showing a weak negative correlation in the HV–HV mode, while exhibiting a significant positive correlation in the HV–AV mode. Notably, the correlations vary across different driving styles and driving regimes. For instance, in the HV–HV mode, the correlation between the pair (v_m, a_m) is negative in regime #2 (HS) but positive in regime #3 (MS), with coefficients of -0.10 and 0.04 , respectively. Similarly, in driving regime #1 (LS), the correlation between the pair (s_0, b_m) is significantly negative for both the neutral and timid styles (with coefficients of -0.21 and -0.20), yet minimal for the aggressive style, where the coefficient is -0.07 .

Furthermore, from the correlation matrices of the neutral driving styles for the Lyft dataset (see Fig. B.14), the IDM parameter correlations in the Lyft dataset are quite similar to those in the Waymo dataset, though certain differences also exist. For instance, the pair (v_m, s_0) exhibits a strong negative correlation in the Waymo dataset, whereas it exhibits a slight negative correlation in the Lyft dataset. The reason may lie in the differences in traffic conditions: the Waymo dataset represents a freeway scenario, whereas the Lyft dataset corresponds to an urban road scenario. These findings provide evidence that human drivers with the same driving style exhibit varied driving behaviors across various car-following modes and driving regimes, while drivers with distinct styles may also display unique behaviors within the same regime. Additionally, the correlation matrices of the IDM parameter θ , across different driving regimes are consistent when using both the hierarchical and unpooled models.

5.4. Simulation evaluation

Deterministic simulation is widely used in model calibration using trajectory data. In particular, given the motion information of the leading vehicle and the initial states (e.g., initial speed, position, etc.) of the following vehicle, the simulation can generate the simulated motion states using the suggested model parameters. The performance is generated by comparing the simulated motion states with the observed data. According to the intrinsic probabilistic nature of Bayesian calibration, Zhang et al. (2024) proposed a stochastic simulation approach. The key difference between deterministic and stochastic simulations lies in the approach to generating the action $a^{(t)}$. In stochastic simulation, the action $a^{(t)}$ is sampled from Eq. (10), i.e., $a^{(t)} = a_{\text{IDM}}^{(t)} + \epsilon_t$, whereas deterministic simulation utilizes a fixed action, i.e., $a^{(t)} = a_{\text{IDM}}^{(t)}$.

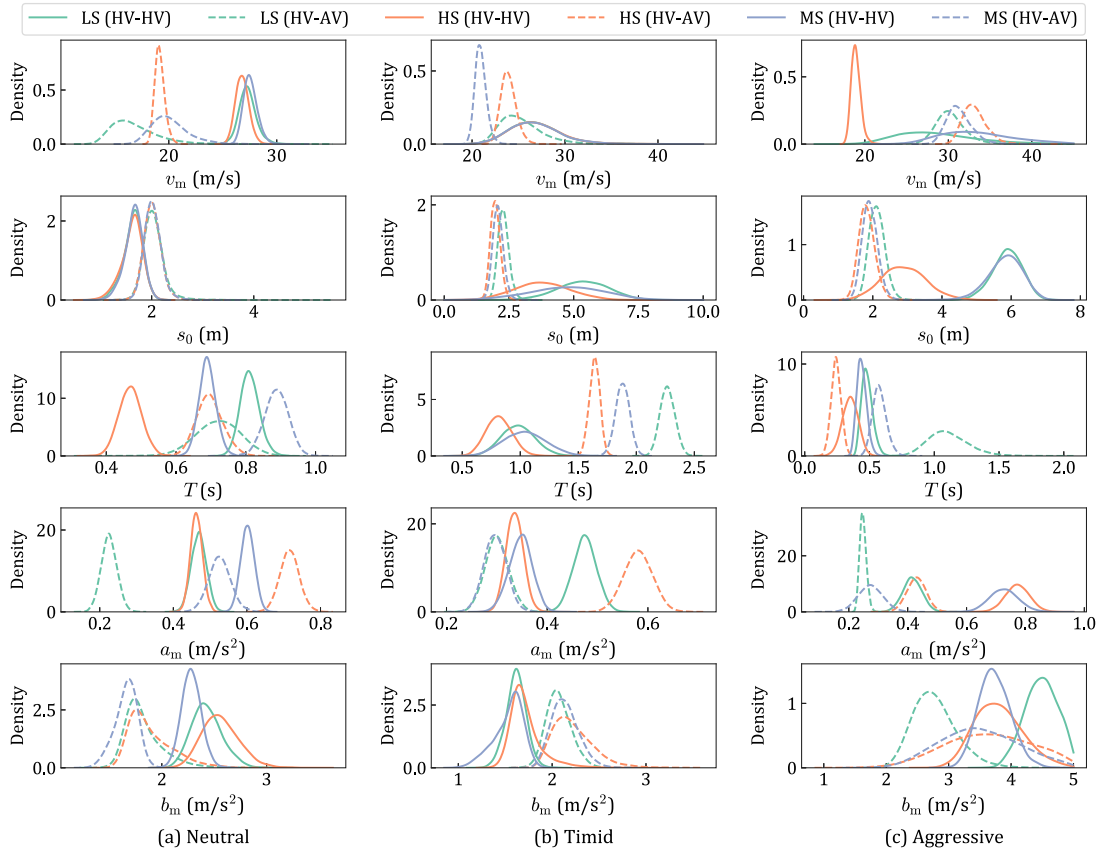


Fig. 8. Kernel density estimations of model parameters for hierarchical B-IDM for the Waymo dataset.

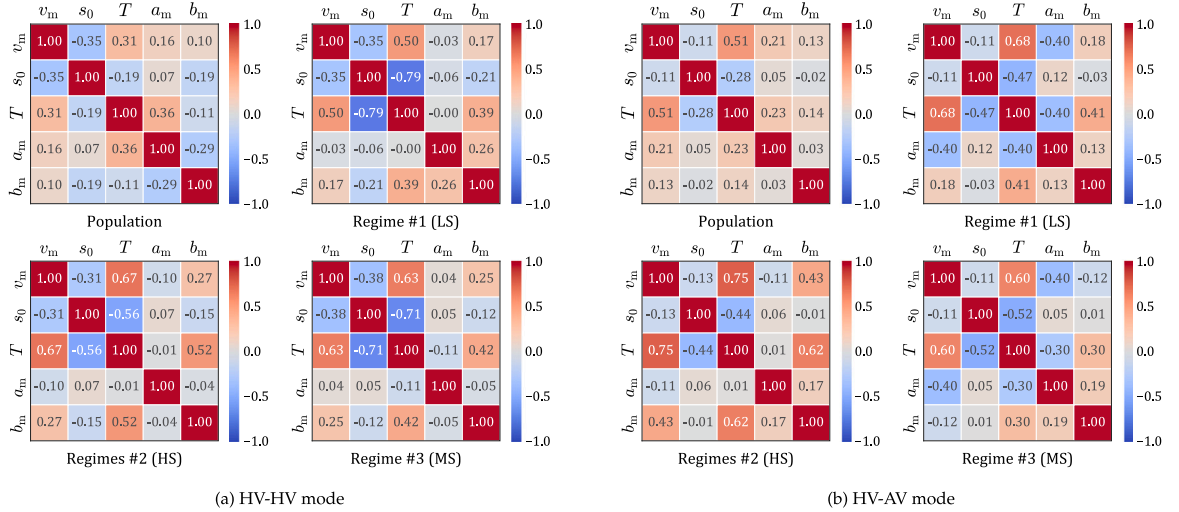


Fig. 9. Correlation matrices of hierarchical B-IDM parameters under the neutral driving style for the Waymo dataset.

Previous studies have demonstrated that drawing model parameters solely from uncorrelated marginal distributions can result in unreliable behaviors in microscopic simulations (Kim and Mahmassani, 2011; Wu et al., 2024). Thanks to the Bayesian calibration approach, joint distributions of IDM parameters are obtained, rather than point estimates derived from traditional optimization-based methods. Therefore, instead of using the expectation values from Table 3, enormous model parameters are drawn from the joint posterior distribution to generate diverse driving behaviors. In this study, a representative car-following event is selected for

Table 4
Summary of simulation results with different models for the Waymo dataset.

Mode	Style	Model	Deterministic			Stochastic		
			RMSE (<i>a</i>)	RMSE (<i>v</i>)	RMSE (<i>s</i>)	CRPS (<i>a</i>)	CRPS (<i>v</i>)	CRPS (<i>s</i>)
HV–HV	Neutral (ID: 753)	Naive IDM	0.265	0.570	1.326	–	–	–
		PB-IDM	0.252 ± 0.001	0.443 ± 0.003	2.458 ± 0.080	0.147 ± 0.072	0.328 ± 0.233	1.969 ± 1.125
		HB-IDM	0.243 ± 0.005	0.406 ± 0.017	0.858 ± 0.052	0.141 ± 0.066	0.300 ± 0.197	0.517 ± 0.445
		UB-IDM	0.251 ± 0.003	0.410 ± 0.004	2.098 ± 0.059	0.147 ± 0.097	0.307 ± 0.223	1.627 ± 1.095
	Timid (ID: 1653)	Naive IDM	0.293	0.330	0.672	–	–	–
		PB-IDM	0.217 ± 0.000	0.328 ± 0.003	2.027 ± 0.066	0.126 ± 0.063	0.194 ± 0.219	1.561 ± 0.976
		HB-IDM	0.208 ± 0.006	0.325 ± 0.010	0.645 ± 0.099	0.118 ± 0.100	0.185 ± 0.152	0.353 ± 0.254
		UB-IDM	0.218 ± 0.002	0.351 ± 0.004	1.942 ± 0.045	0.128 ± 0.086	0.194 ± 0.243	1.341 ± 0.996
	Aggressive (ID: 48)	Naive IDM	0.514	0.765	2.326	–	–	–
		PB-IDM	0.372 ± 0.002	0.544 ± 0.005	1.680 ± 0.091	0.205 ± 0.168	0.340 ± 0.376	1.163 ± 1.068
		HB-IDM	0.365 ± 0.007	0.535 ± 0.025	1.530 ± 0.185	0.197 ± 0.175	0.325 ± 0.355	0.864 ± 1.071
		UB-IDM	0.371 ± 0.004	0.540 ± 0.005	1.669 ± 0.078	0.202 ± 0.205	0.337 ± 0.410	1.035 ± 1.217
HV–AV	Neutral (ID: 1941)	Naive IDM	0.401	0.763	2.085	–	–	–
		PB-IDM	0.399 ± 0.003	0.816 ± 0.017	3.561 ± 0.129	0.234 ± 0.168	0.577 ± 0.490	2.242 ± 2.355
		HB-IDM	0.350 ± 0.009	0.675 ± 0.028	2.119 ± 0.246	0.203 ± 0.127	0.461 ± 0.404	1.303 ± 1.132
		UB-IDM	0.389 ± 0.003	0.745 ± 0.013	3.430 ± 0.140	0.222 ± 0.173	0.518 ± 0.479	2.314 ± 2.514
	Timid (ID: 1151)	Naive IDM	0.273	0.481	1.463	–	–	–
		PB-IDM	0.343 ± 0.001	0.685 ± 0.004	3.451 ± 0.062	0.200 ± 0.109	0.494 ± 0.400	2.516 ± 2.016
		HB-IDM	0.282 ± 0.005	0.407 ± 0.015	1.350 ± 0.153	0.164 ± 0.085	0.268 ± 0.217	0.765 ± 0.694
		UB-IDM	0.336 ± 0.002	0.645 ± 0.006	3.480 ± 0.050	0.195 ± 0.123	0.488 ± 0.360	2.316 ± 2.040
	Aggressive (ID: 759)	Naive IDM	0.539	0.665	1.966	–	–	–
		PB-IDM	0.685 ± 0.004	0.768 ± 0.004	2.328 ± 0.032	0.339 ± 0.426	0.540 ± 0.474	1.587 ± 1.600
		HB-IDM	0.408 ± 0.008	0.440 ± 0.013	1.080 ± 0.078	0.252 ± 0.126	0.298 ± 0.214	0.747 ± 0.493
		UB-IDM	0.578 ± 0.004	0.607 ± 0.006	2.332 ± 0.033	0.292 ± 0.362	0.420 ± 0.392	1.566 ± 1.025

Note: the best results are highlighted in bold. PB-IDM, HB-IDM, and UB-IDM denote the pooled, hierarchical, and unpooled B-IDM.

Table 5
Summary of simulation results with different models for the Lyft dataset.

Mode	Style	Model	Deterministic			Stochastic		
			RMSE (<i>a</i>)	RMSE (<i>v</i>)	RMSE (<i>s</i>)	CRPS (<i>a</i>)	CRPS (<i>v</i>)	CRPS (<i>s</i>)
HH	Neutral (ID: 3)	Naive IDM	0.104	0.252	1.768	–	–	–
		PB-IDM	0.121 ± 0.002	0.293 ± 0.005	1.779 ± 0.091	0.087 ± 0.044	0.228 ± 0.096	1.379 ± 0.875
		HB-IDM	0.101 ± 0.004	0.178 ± 0.018	0.836 ± 0.245	0.083 ± 0.024	0.104 ± 0.073	0.494 ± 0.289
		UB-IDM	0.165 ± 0.005	0.335 ± 0.010	2.066 ± 0.083	0.093 ± 0.072	0.257 ± 0.163	1.698 ± 0.957
	Timid (ID: 663)	Naive IDM	0.244	0.496	1.942	–	–	–
		PB-IDM	0.286 ± 0.001	0.559 ± 0.004	1.483 ± 0.109	0.161 ± 0.123	0.364 ± 0.366	0.969 ± 0.766
		HB-IDM	0.225 ± 0.004	0.409 ± 0.010	0.850 ± 0.258	0.130 ± 0.079	0.224 ± 0.271	0.409 ± 0.290
		UB-IDM	0.288 ± 0.002	0.557 ± 0.009	2.750 ± 0.144	0.165 ± 0.138	0.406 ± 0.320	2.116 ± 1.356
	Aggressive (ID: 289)	Naive IDM	0.498	0.682	1.739	–	–	–
		PB-IDM	0.499 ± 0.000	0.671 ± 0.003	1.863 ± 0.064	0.289 ± 0.224	0.521 ± 0.324	1.204 ± 1.219
		HB-IDM	0.491 ± 0.003	0.578 ± 0.007	1.074 ± 0.078	0.287 ± 0.213	0.414 ± 0.293	0.601 ± 0.470
		UB-IDM	0.498 ± 0.001	0.640 ± 0.003	1.635 ± 0.045	0.301 ± 0.257	0.504 ± 0.326	1.136 ± 1.043
HA	Neutral (ID: 1007)	Naive IDM	0.192	0.396	1.848	–	–	–
		PB-IDM	0.172 ± 0.000	0.248 ± 0.006	1.706 ± 0.073	0.106 ± 0.052	0.155 ± 0.150	1.359 ± 0.773
		HB-IDM	0.171 ± 0.003	0.232 ± 0.029	0.972 ± 0.217	0.098 ± 0.070	0.151 ± 0.160	0.607 ± 0.458
		UB-IDM	0.175 ± 0.001	0.360 ± 0.006	3.049 ± 0.050	0.100 ± 0.061	0.268 ± 0.197	2.649 ± 1.300
	Timid (ID: 1905)	Naive IDM	0.217	0.307	1.262	–	–	–
		PB-IDM	0.258 ± 0.001	0.534 ± 0.005	1.232 ± 0.024	0.150 ± 0.071	0.408 ± 0.266	0.853 ± 0.621
		HB-IDM	0.208 ± 0.004	0.262 ± 0.011	0.772 ± 0.094	0.123 ± 0.064	0.170 ± 0.133	0.492 ± 0.385
		UB-IDM	0.264 ± 0.002	0.567 ± 0.005	1.367 ± 0.027	0.161 ± 0.086	0.446 ± 0.293	1.003 ± 0.752
	Aggressive (ID: 1531)	Naive IDM	0.389	0.685	1.634	–	–	–
		PB-IDM	0.442 ± 0.001	0.888 ± 0.008	2.709 ± 0.055	0.259 ± 0.161	0.675 ± 0.493	2.133 ± 1.357
		HB-IDM	0.340 ± 0.006	0.525 ± 0.019	1.103 ± 0.099	0.195 ± 0.112	0.322 ± 0.342	0.592 ± 0.584
		UB-IDM	0.415 ± 0.002	0.795 ± 0.009	2.809 ± 0.056	0.256 ± 0.156	0.618 ± 0.428	2.124 ± 1.636

Note: the best results are highlighted in bold. PB-IDM, HB-IDM, and UB-IDM denote the pooled, hierarchical, and unpooled B-IDM.

each combination of car-following mode and driving style, yielding a total of 12 events for simulation. For the Waymo dataset, the car-following IDs 753, 1653, and 48 represent the neutral, timid, and aggressive styles, respectively, in HV–HV mode, while IDs 1941, 1151, and 759 correspond to the same driving styles in HV–AV mode. Similarly, in the Lyft dataset, IDs 3, 663, and 289 are associated with HV–HV mode, and IDs 1007, 1950, and 1531 with HV–AV mode. A total of $N = 1000$ sets of IDM parameters are

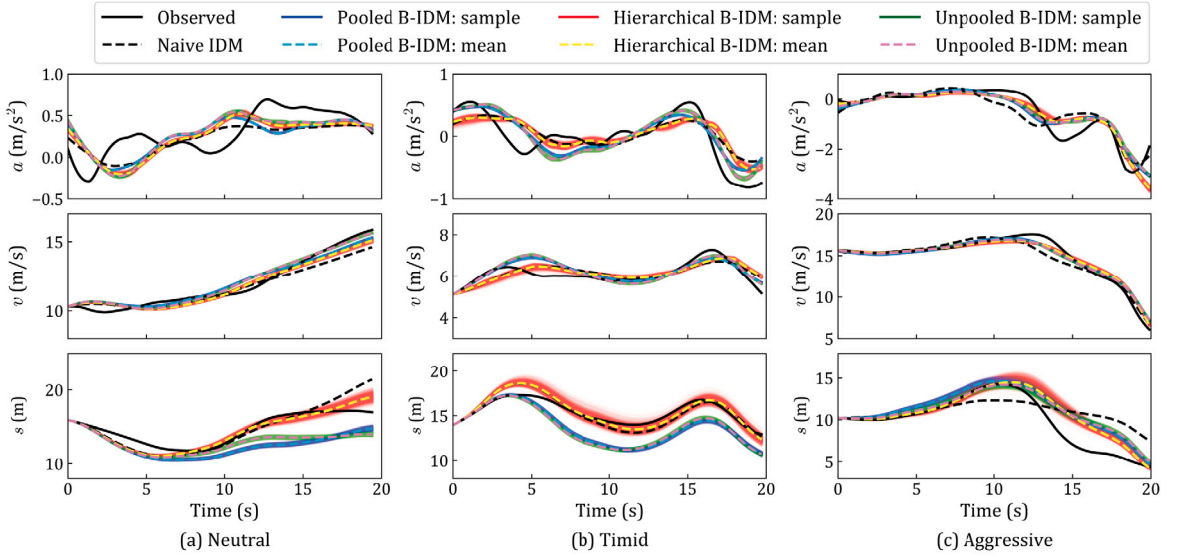


Fig. 10. Simulated motion state plots from deterministic simulations for the HV-HV mode for the Waymo dataset.

sampled for each model. Given the trajectory of the leading vehicle and the same initial motion states of the following vehicle, the trajectories of the following vehicle are simulated using the sampled IDM parameters. Note that in deterministic simulation, the action is directly obtained from IDM, that is $a^{(t)} = a_{\text{IDM}}^{(t)}$. In contrast, for stochastic simulation, the action is given by $a^{(t)} = a_{\text{IDM}}^{(t)} + \epsilon_t$, where ϵ_t is sampled from $\mathcal{N}(0, \sigma_\epsilon^2)$ and σ_ϵ represents the noise level.

To assess simulation accuracy, the root mean square error (RMSE) of motion states (i.e., a , v , and s) is employed to evaluate the performance of deterministic simulations, whereas the continuous ranked probability score (CRPS) (Gneiting and Raftery, 2007; Chen et al., 2023) is used to assess the performance of stochastic simulations.

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{t=1}^T (y_t - \hat{y}_t)^2} \quad (28)$$

$$\text{CRPS}(y_t) = \int_{-\infty}^{\infty} (F(y) - \mathbb{I}\{y > y_t\})^2 dy \quad (29)$$

where y and $\hat{y}$ are the observed and simulated motion states, respectively. y_t is the observation at time t , F denotes the forecasting cumulative distribution functions, and $\mathbb{I}$ represents the indicator function. The simulation results are respectively summarized in Tables 4 and 5 in the form of (mean $\pm$ standard deviation), with models exhibiting the lowest errors highlighted in bold. Note that this study also performs the deterministic simulation using the naive IDM, with the model parameters calibrated through a genetic algorithm (as shown in Appendix C). The simulation results are evaluated using RMSE, and the mean values are summarized in Tables 4 and 5.

It is evident that the pooled B-IDM exhibits the worst performance in both deterministic and stochastic simulations for the Waymo dataset, whereas the unpooled B-IDM demonstrates approximately the highest errors in the Lyft dataset. Remember that the pooled model learns general driving behaviors by fitting all car-following data with diverse driving regimes. However, driving behaviors may vary significantly across diverse driving regimes. The unpooled model develops distinct models for each driving regime but fails to account for the correlations between these regimes. These are the reasons for the limited performance of the pooled and unpooled models, respectively. The hierarchical B-IDM achieves the best simulation results, with the lowest errors. Additionally, the hierarchical B-IDM outperforms the naive IDM, which fails to capture driving behaviors across different driving regimes. Taking the neutral style in the HV-HV mode of the Waymo dataset as an example (see Table 4), the average errors in the deterministic simulation are 0.243 for acceleration, 0.406 for speed, and 0.858 for spacing. In stochastic simulation, the corresponding errors are 0.141, 0.300, and 0.517, respectively. Figs. 10 and 11 illustrate the observed and simulated motion states in the HV-HV mode for the Waymo dataset. Similarly, the HV-HV mode for the Waymo dataset and both car-following modes for the Lyft dataset are shown in Figs. D.15–D.20. In deterministic simulations, the simulated motion states produced by the hierarchical B-IDM are closer to the observed values than those generated by the naive IDM, pooled B-IDM, and unpooled B-IDM. Similarly, in stochastic simulations, the hierarchical B-IDM outperforms both the pooled and unpooled B-IDM. These support the findings that hierarchical models can calibrate disparate IDM parameters across diverse driving regimes, thereby improving the simulation accuracy. The unpooled B-IDM also accounts for diverse model parameters but neglects the correlations between different driving regimes. As a result, the simulation performance of the unpooled B-IDM is inferior to that of the hierarchical B-IDM.

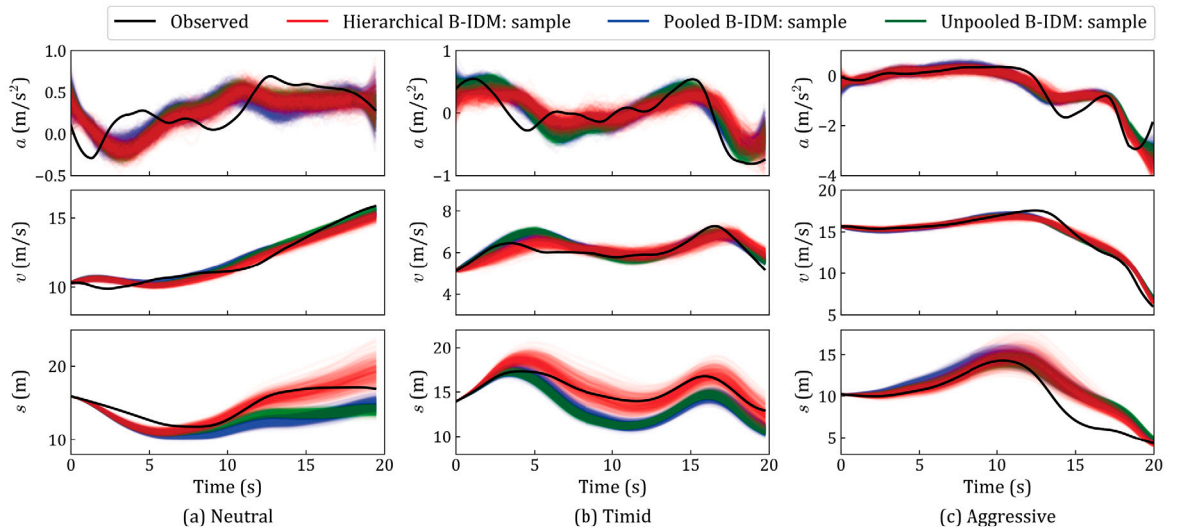


Fig. 11. Simulated motion state plots from stochastic simulations for the HV-HV mode for the Waymo dataset.

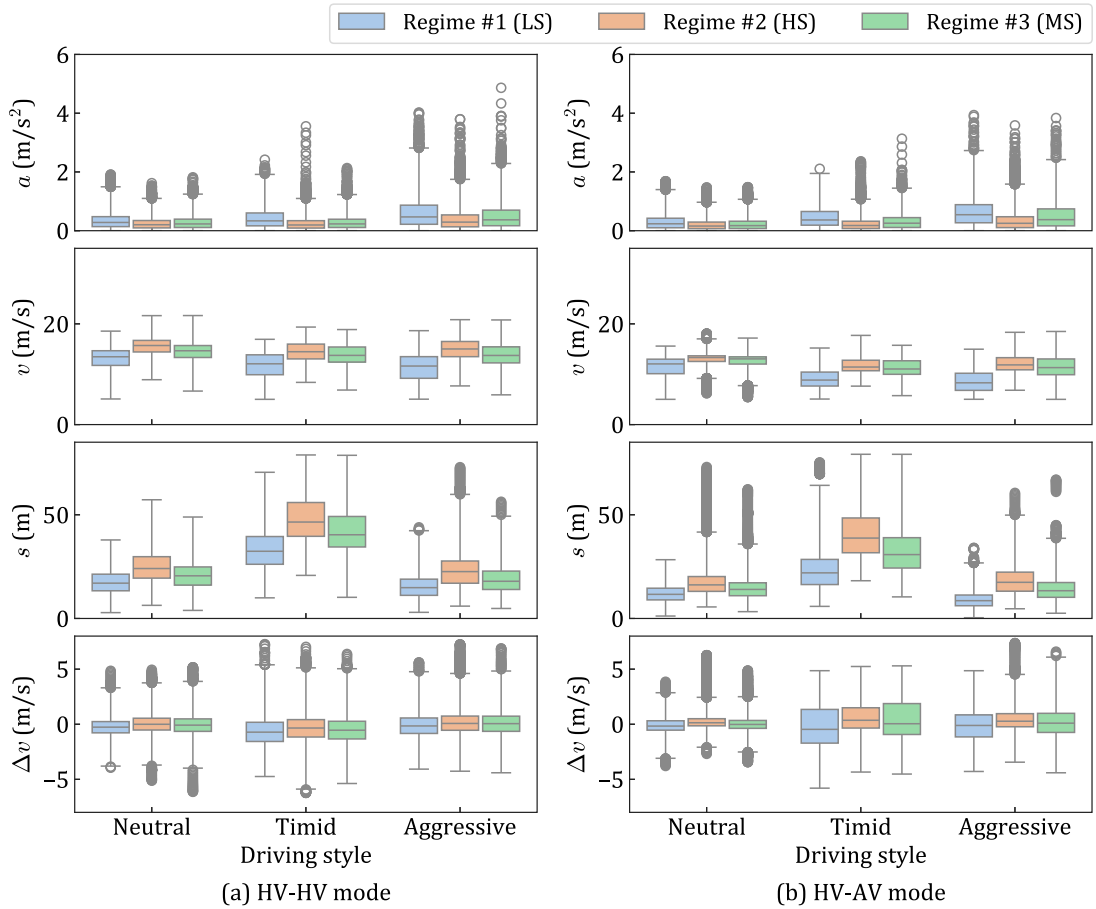


Fig. A.12. Boxplots of driving parameters for the Lyft dataset.

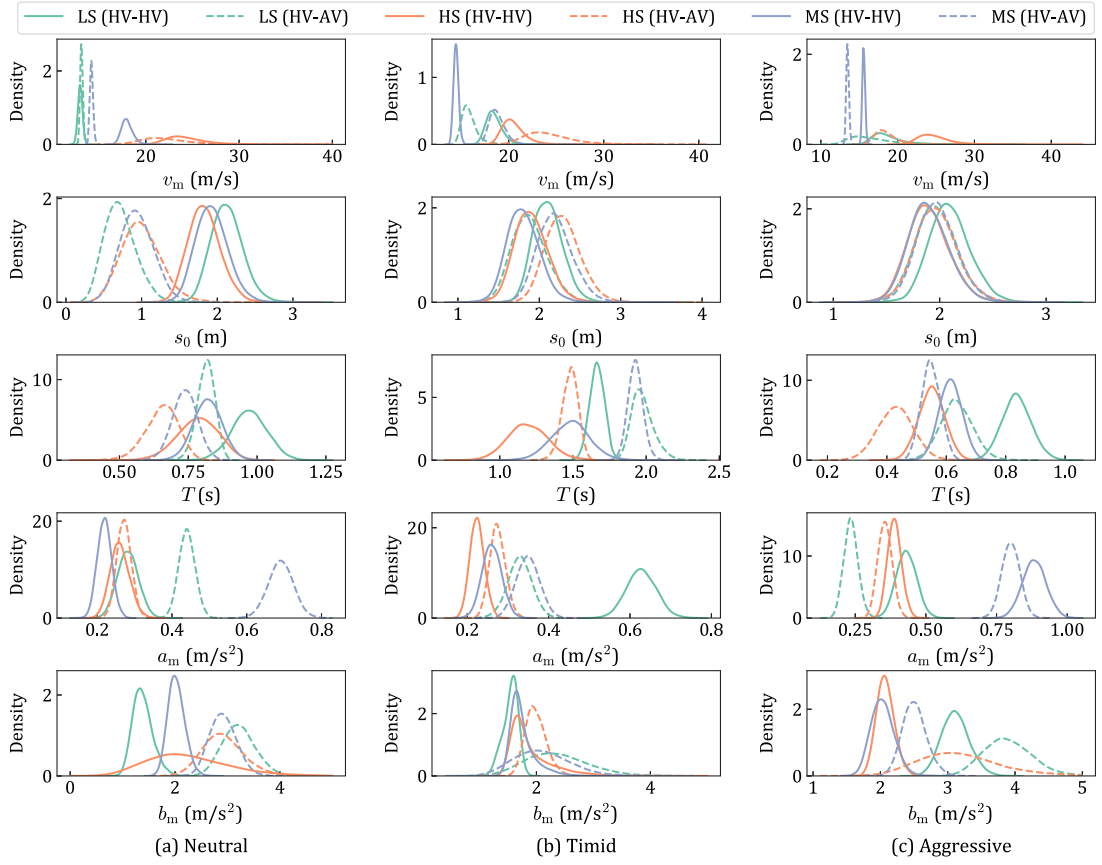


Fig. B.13. Kernel density estimations of model parameters for hierarchical B-IDM for the Lyft dataset.

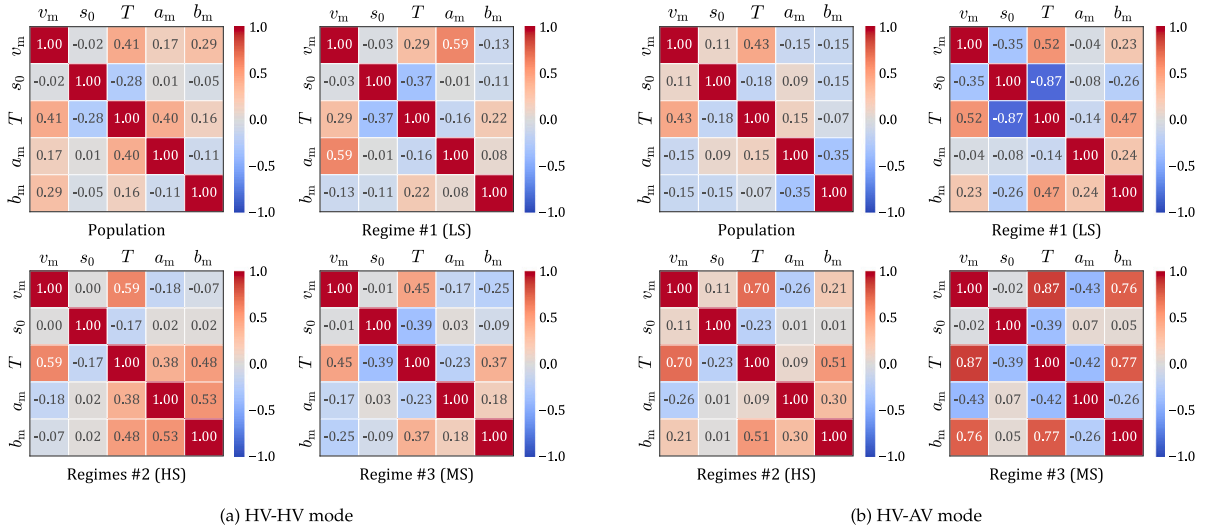


Fig. B.14. Correlation matrices of hierarchical B-IDM parameters under the neutral driving style for the Waymo dataset.

Table B.6

Posterior expectations of parameters for pooled, hierarchical, and unpooled B-IDM in the Lyft dataset.

Style	Model	Regime	HV–HV mode		HV–AV mode	
			$\theta = [v_m, s_0, T, a_m, b_m]$	σ_ϵ	$\theta = [v_m, s_0, T, a_m, b_m]$	σ_ϵ
Neutral	PB-IDM	Pop	[18.660, 1.989, 0.964, 0.645, 2.840]	0.292	[17.434, 0.927, 0.825, 0.554, 2.206]	0.288
		LS	[12.962, 2.130, 0.972, 0.282, 1.384]	0.366	[13.101, 0.719, 0.814, 0.440, 3.219]	0.223
	HB-IDM	Pop	[18.824, 2.029, 0.937, 0.358, 2.154]	0.113	[17.277, 1.002, 0.773, 0.553, 3.302]	0.095
		LS	[12.962, 2.130, 0.972, 0.282, 1.384]	0.366	[13.101, 0.719, 0.814, 0.440, 3.219]	0.223
		HS	[24.293, 1.831, 0.785, 0.261, 2.254]	0.243	[22.130, 0.996, 0.654, 0.272, 2.921]	0.206
		MS	[17.999, 1.929, 0.821, 0.219, 2.026]	0.272	[14.173, 0.940, 0.738, 0.691, 2.916]	0.202
	UB-IDM	LS	[14.487, 1.987, 0.888, 0.699, 1.832]	0.239	[17.211, 0.946, 0.848, 0.568, 2.119]	0.310
		HS	[22.307, 1.996, 1.103, 0.682, 3.510]	0.179	[25.331, 1.444, 1.096, 0.657, 1.852]	0.199
		MS	[17.654, 1.986, 0.799, 0.662, 2.452]	0.203	[21.925, 1.291, 1.025, 0.648, 1.966]	0.208
Timid	PB-IDM	Pop	[16.751, 1.997, 1.588, 0.669, 1.676]	0.277	[16.575, 1.978, 1.486, 0.626, 1.836]	0.253
	HB-IDM	Pop	[18.616, 2.005, 1.457, 0.451, 1.691]	0.112	[20.102, 2.082, 1.716, 0.427, 2.886]	0.104
		LS	[18.489, 2.107, 1.665, 0.628, 1.546]	0.370	[15.914, 1.885, 1.976, 0.329, 2.367]	0.310
		HS	[20.518, 1.901, 1.192, 0.225, 1.889]	0.212	[24.207, 2.292, 1.480, 0.275, 1.989]	0.192
		MS	[14.457, 1.807, 1.483, 0.261, 1.750]	0.300	[18.806, 2.195, 1.927, 0.349, 2.101]	0.250
	UB-IDM	LS	[16.458, 1.994, 1.600, 0.734, 1.646]	0.227	[14.268, 1.965, 1.451, 0.667, 1.784]	0.227
		HS	[16.963, 1.991, 1.187, 0.659, 1.684]	0.169	[18.332, 2.003, 1.617, 0.660, 1.738]	0.157
		MS	[14.919, 1.993, 1.288, 0.680, 1.676]	0.215	[16.498, 1.980, 1.481, 0.680, 1.775]	0.191
	Aggressive	PB-IDM	Pop	[20.390, 1.970, 0.824, 0.693, 2.794]	0.373	[14.473, 1.453, 0.537, 0.634, 2.382]
HB-IDM		Pop	[20.021, 1.996, 0.713, 0.646, 2.224]	0.139	[17.152, 1.969, 0.531, 0.528, 3.167]	0.130
		LS	[18.778, 2.094, 0.837, 0.431, 3.114]	0.480	[16.326, 1.961, 0.635, 0.235, 3.898]	0.423
		HS	[24.769, 1.896, 0.551, 0.389, 2.074]	0.362	[18.538, 1.966, 0.430, 0.355, 3.179]	0.372
		MS	[15.582, 1.898, 0.614, 0.884, 2.032]	0.489	[13.482, 1.976, 0.548, 0.801, 2.505]	0.355
UB-IDM		LS	[17.346, 1.964, 0.896, 0.700, 3.330]	0.282	[11.891, 1.397, 0.540, 0.622, 2.142]	0.298
		HS	[22.747, 1.971, 0.808, 0.690, 2.339]	0.240	[15.574, 1.722, 0.716, 0.677, 2.255]	0.293
		MS	[18.168, 1.964, 0.759, 0.728, 2.342]	0.300	[14.194, 1.553, 0.571, 0.691, 2.328]	0.254

Note: PB-IDM, HB-IDM, and UB-IDM denote the pooled, hierarchical, and unpooled B-IDM, and Pop denotes the population level. The recommendation value: $\theta_{\text{rec}} = [25.0, 2.0, 1.6, 1.5, 1.67]$.

Table C.7

Calibration results of naive IDM.

Mode	Style	Waymo	Lyft
		$\theta = [v_m, s_0, T, a_m, b_m]$	$\theta = [v_m, s_0, T, a_m, b_m]$
HV–HV	Neutral	[28.897, 1.368, 0.742, 0.426, 2.823]	[20.146, 0.303, 1.260, 0.215, 2.095]
	Timid	[22.023, 8.268, 0.886, 0.464, 2.047]	[17.390, 1.290, 1.596, 0.273, 1.452]
	Aggressive	[30.840, 3.090, 0.570, 0.742, 0.407]	[20.113, 1.019, 0.819, 0.551, 3.034]
HV–AV	Neutral	[29.811, 2.510, 0.723, 0.388, 1.414]	[15.799, 1.910, 0.695, 0.225, 4.827]
	Timid	[23.442, 6.971, 1.308, 0.493, 2.689]	[19.889, 2.239, 1.816, 0.244, 3.782]
	Aggressive	[23.542, 2.519, 0.311, 0.158, 4.990]	[19.548, 1.319, 0.551, 0.330, 3.101]

6. Conclusions

To effectively capture driver heterogeneity and provide a precise representation of dynamic human car-following behaviors, this study proposes a stochastic modeling framework that integrates multi-regime car-following models with Bayesian calibration to account for time-varying parameters. The Waymo dataset and Lyft dataset are utilized for analysis. According to the results, the main conclusions are summarized as follows: (1) The K-means clustering algorithm groups human drivers into distinct driving styles to capture inter-driver heterogeneity, while the HDP-HSMM segments driving regimes to account for intra-driver heterogeneity. These findings reveal significant differences in human driving behaviors across various driving styles and driving regimes. The results suggest the necessity of modeling heterogeneous human driving behaviors to capture these variations accurately. (2) The Bayesian approach effectively calibrates multi-regime pooled, hierarchical, and unpooled B-IDM, particularly for the hierarchical B-IDM, enabling the representation of time-varying parameters to describe heterogeneous driving behaviors accurately. (3) The deterministic and stochastic simulations were conducted, with the hierarchical B-IDM achieving the best performance. Also, the stochastic simulation outperforms the deterministic simulation in replicating human driving behavior. Overall, this study provides valuable insights into modeling heterogeneous driving behaviors and understanding car-following dynamics by jointly accounting for inter- and intra-driver heterogeneity. These insights are essential for enhancing the behavioral realism and adaptability of traffic models, and they hold strong potential for informing the development of human-like decision-making strategies in autonomous vehicles and advanced driver-assistance systems.

Although the proposed stochastic modeling framework effectively captures both inter-driver and intra-driver heterogeneity and provides an accurate representation of heterogeneous human driving behaviors, several aspects can be further explored in future

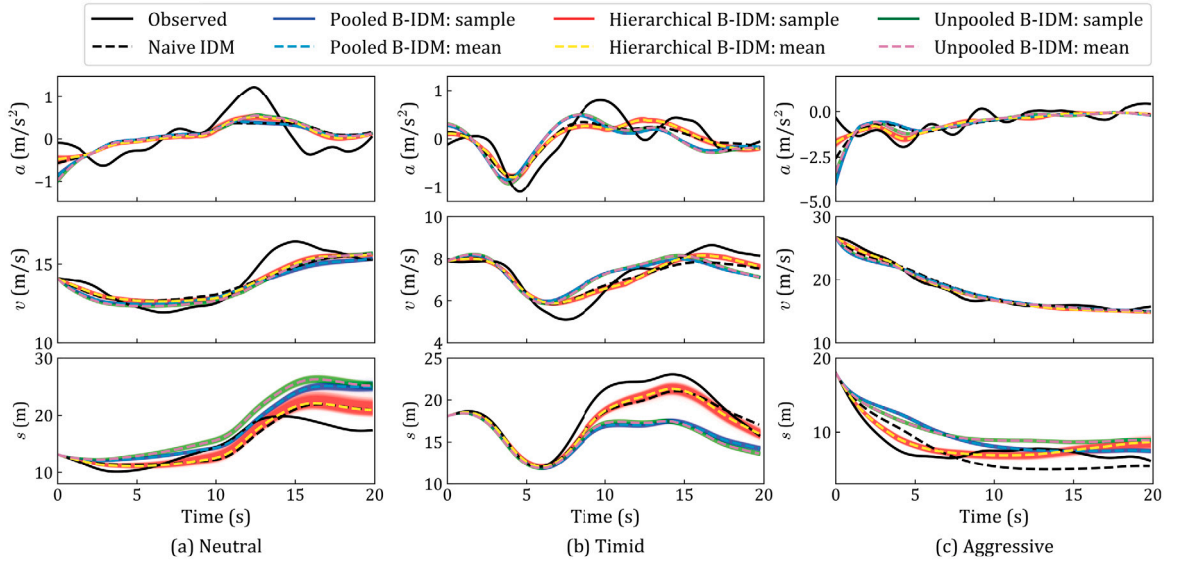


Fig. D.15. Simulated motion state plots from deterministic simulations for the HV-AV mode for the Waymo dataset.

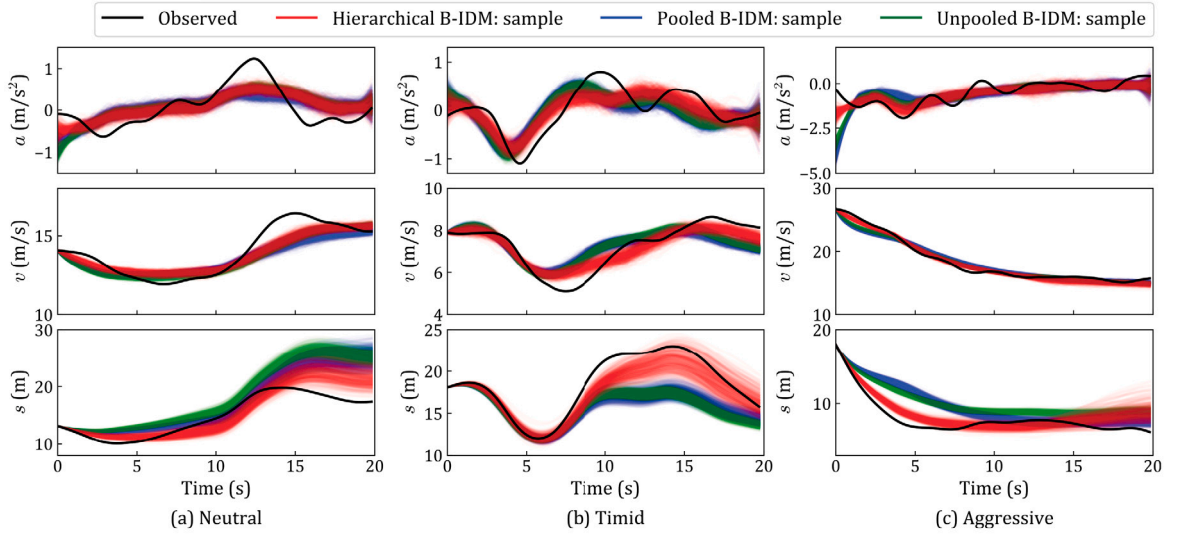


Fig. D.16. Simulated motion state plots from stochastic simulations for the HV-AV mode for the Waymo dataset.

work. (1) More realistic driving scenarios involve multiple factors, as a driver's decisions during car-following may be influenced by surrounding vehicles, especially in congested conditions. Therefore, the segmentation of driving regimes should incorporate information from not only the leading and subject vehicles but also surrounding traffic to capture the full complexity of driving behavior. (2) In this study, acceleration noise σ_e is treated as uncorrelated to maintain model traceability. In future work, we intend to incorporate temporally correlated acceleration noise by applying time series modeling techniques (e.g., autoregressive moving average) to capture its temporal structure. (3) This study selects the IDM as a representative example for analysis. Future work will incorporate additional car-following models (e.g., the Gipps model) to further validate the generality of the proposed stochastic framework for modeling heterogeneous car-following behavior.

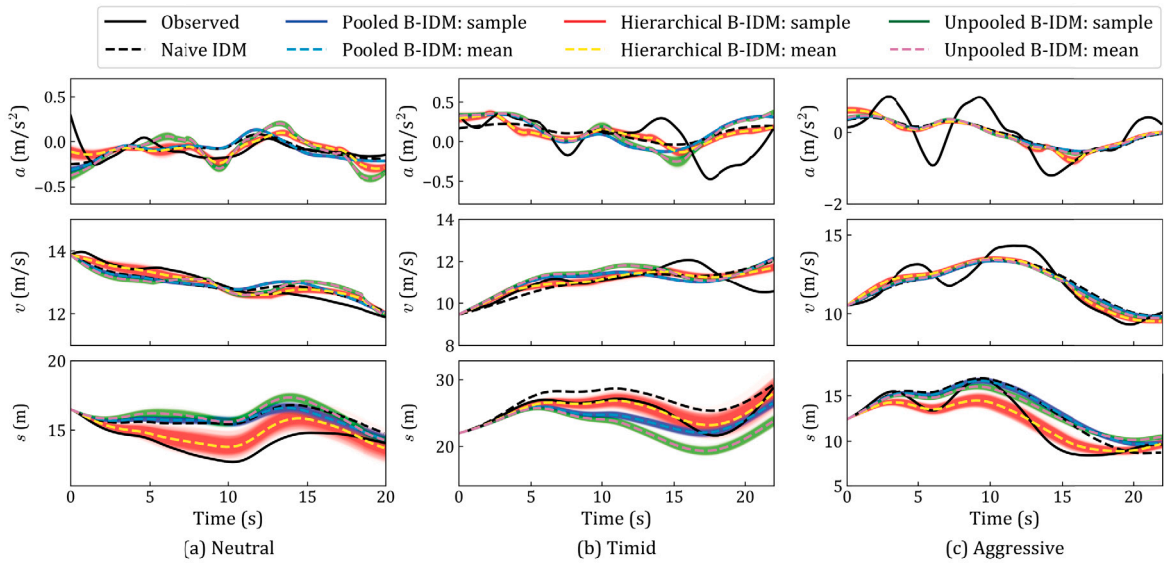


Fig. D.17. Simulated motion state plots from deterministic simulations for the HV-HV mode for the Lyft dataset.

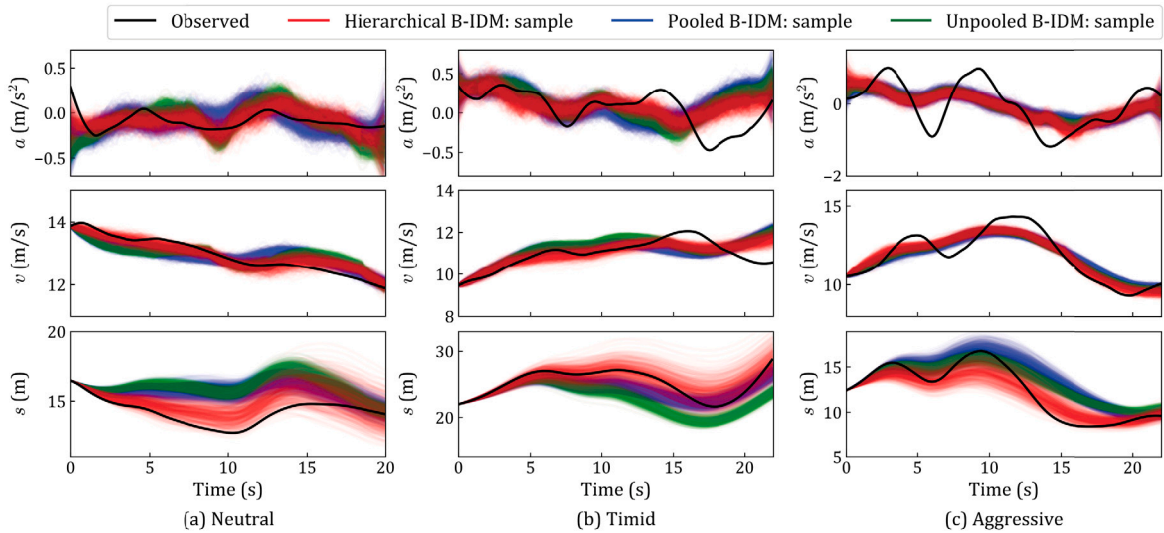


Fig. D.18. Simulated motion state plots from stochastic simulations for the HV-HV mode for the Lyft dataset.

CRediT authorship contribution statement

Shubo Wu: Methodology, Data curation, Formal analysis, Writing – review & editing, Writing – original draft. **Dong Ngoduy:** Writing – review & editing, Formal analysis, Writing – original draft. **Zhengbing He:** Writing – review & editing, Investigation, Formal analysis. **Yajie Zou:** Funding acquisition, Writing – review & editing, Formal analysis, Writing – original draft, Supervision. **Jian Sun:** Methodology, Formal analysis, Writing – review & editing.

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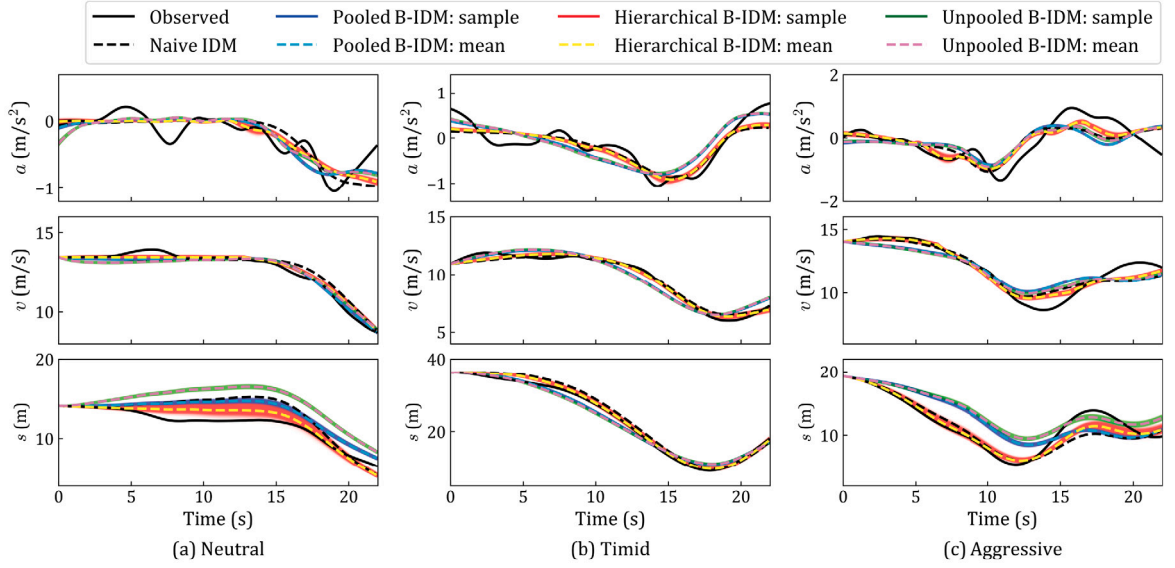


Fig. D.19. Simulated motion state plots from deterministic simulations for the HV-AV mode for the Lyft dataset.

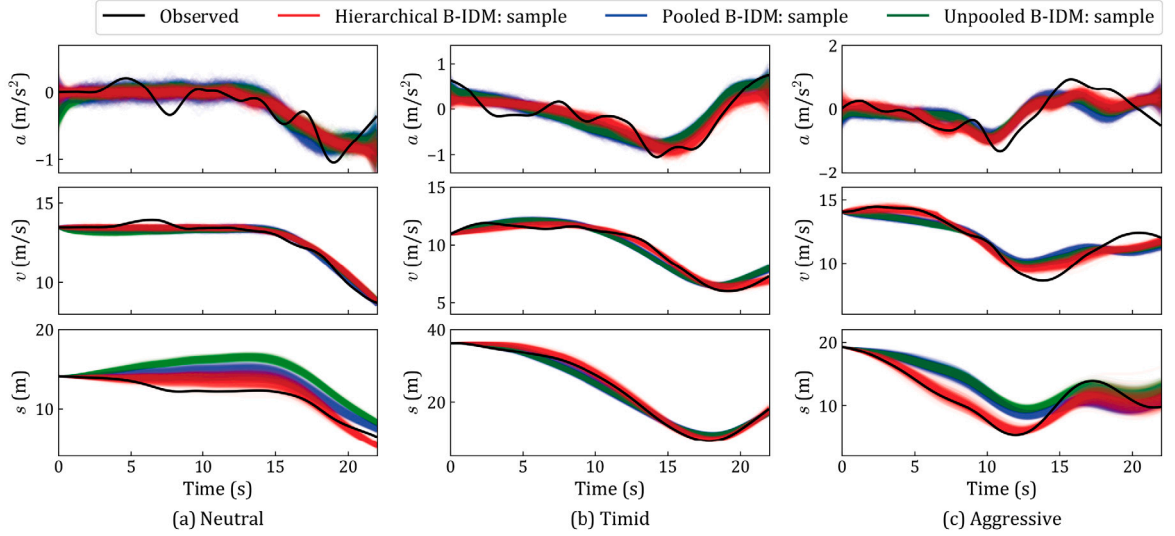


Fig. D.20. Simulated motion state plots from stochastic simulations for the HV-AV mode for the Lyft dataset.

Appendix A. The boxplots of driving parameters for the Lyft dataset

Here, the boxplots of driving parameters (i.e., acceleration, spacing, speed, and speed difference) are illustrated in Fig. A.12.

Appendix B. Calibration results for the Lyft dataset

Here, the calibration results for the Lyft dataset are provided in Table B.6, the kernel density estimations of hierarchical B-IDM are parameters illustrated in Fig. B.13, and the correlation matrices of parameters under the neutral driving style for the hierarchical B-IDM are shown in Fig. B.14.

Appendix C. Calibration of naive IDM

The parameters of naive IDM are calibrated with a genetic algorithm. The objective function is designed using normalized root square error (NRMSE) with speed v being the measure of performance (see (C.1)). In addition, the parameter bounds of IDM are given according to the corresponding physical meanings and previous studies (Zhu et al., 2018a; Punzo et al., 2021). The maximum desired speed v_m is set to [1, 35] m/s, the minimum safe distance s_0 is set to [0.1, 10] m, the desired safe time headway is set to [0.1, 5] s, the maximum desired acceleration a_m and comfortable deceleration b_m are both set to [0.1, 5] m/s². For the parameters of the genetic algorithm, the population size is set to 100, the maximum generation is set to 300, and the convergence tolerance is set to 10^{-6} . The calibration results are summarized in Table C.7.

$$Obj = \sqrt{\frac{\sum_{t=1}^T [v^{\text{obs}}(t) - v^{\text{sim}}(t)]^2}{\sum_{t=1}^T [v^{\text{obs}}(t)]^2}} \quad (\text{C.1})$$

where v^{obs} and v^{sim} are the observed and simulated speeds, respectively.

Appendix D. Simulated motion state plots of simulations

Here, the simulated motion state plots of deterministic and stochastic simulations for HV–HV mode in the Waymo dataset are illustrated in Figs. D.15 and D.16. Similarly, the HV–HV and HV–AV modes for the Lyft dataset are illustrated in Figs. D.17–D.20.

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